ホワイトボックス・スイッチを用いた

In-Network Computing

トヨタ自動車株式会社
コネクティッド先行開発部 InfoTech DCインフラG
プリンシパル・リサーチャー 海老澤 健太郎

2020年11月9日

TOYOTA INFOTECH

MPLS Japan 2020

ホワイトボックス・スイッチが広げる「ネットワークの可能性」

データプレーン プログラミング「P4の動向」

「In-Network Computing の研究動向」

ホワイトボックス・スイッチ「今後への期待」

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ホワイトボックス・スイッチ ユースケース分類

分類	Switching / Routing	Network Functions	Computation
機能例	Layer 2 / Layer 3 Switch Packet Transponder	CGN, Load Balancer, DDoS, INT, Service Chaining, Packet Broker	Data Base (KVS), Aggregation, Consensus, Calculation
実装例	（商用） ArcOS (Arrcus) Cumulus(NVIDIA) OcNOS (IP Infusion) （オープンソース） SONiC RARE (Router for Academia Research & Education) 前半	（商用） NoviFlow Kulcloud （オープンソース） Transparent Security (CableLabs) 論文の参照実装	論文の参照実装

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ネットワークの
新しい可能性

ホワイトボックス・スイッチが広げる「ネットワークの可能性」

データプレーン プログラミング「P4の動向」

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Tofino ASIC 搭載スイッチ（ODM / White Box Switch）

- Accton/Edgecore https://www.edge-core.com/
- Delta Networks, Inc./DNI http://www.dninetworks.com/
- Foxconn/UfiSpace https://www.ufispace.com/
- Interface Masters Technologies https://interfacemasters.com/
- Inventec Corporation https://www.inventec.com/
- STORDIS GmbH https://stordis.com/
- NETBERG https://netbergtw.com/

Tofino ASIC
（ODM / White Box Switch）
は浸透済み（変化無し）

Cisco Silicon One … Router ASIC with P4 support

https://www.cisco.com/c/en/us/solutions/service-provider/innovation/silicon-one.html

One processor. No compromises.

Cisco Silicon One™ Q100 is the industry's first routing silicon to break 10 Tbps with a single ASIC.

2x more the bandwidth

More network bandwidth in a single ASIC than any other routing silicon.

Global routing scale

Ready for massive Internet scale.

3x more performance

More packets per second than any other programmable silicon.

Deep buffering

Protecting against packet loss across congested links.

Programmable

P4 programmable for faster feature delivery and future ready deployments.



Hyperscale cloud operators are also very interested in the Cisco 8000 Series. Microsoft has noted how important the support of SAI (Switch Abstraction Interface) is as it rapidly onboards new systems, using the advanced programmability and scale that Cisco Silicon One delivers:

"Cisco's support for SAI on Cisco 8000 modular routing platforms is a major step forward enabling the SONiC community to combine the latest innovations in silicon and port speed with the density and power efficiencies delivered in chassis-based systems. This greatly contributes to the continuous scale up efforts of cloud infrastructure providers in the face of unceasing traffic growth."
Yousef Khalidi , CVP, Azure Networking, Microsoft Corp

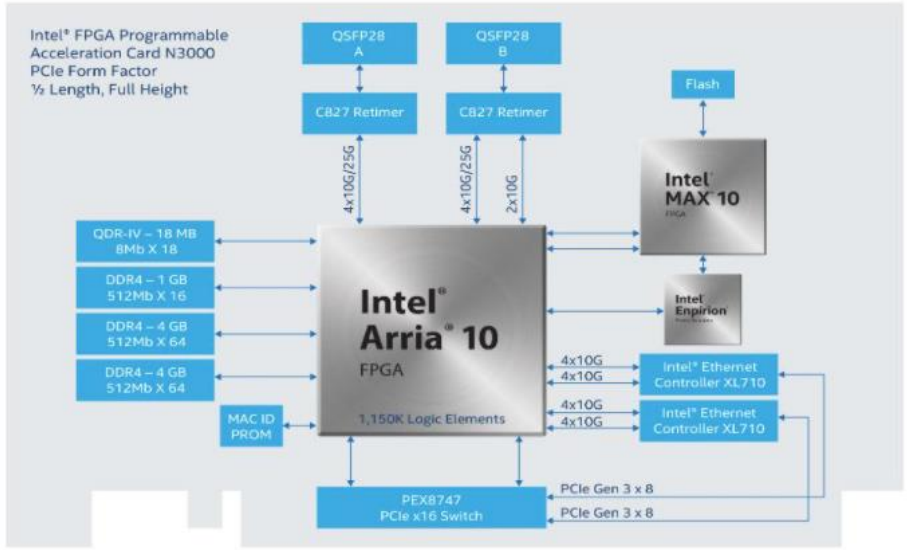
https://blogs.cisco.com/sp/cisco-8000-series-powering-the-internet-for-the-future

どうすればP4を試せるのか？
詳細求む！

Netcope P4 Compiler => Intel P4 Compiler for FPGA

2020年6月に Netcope からインテルが P4 Compiler を部門買収



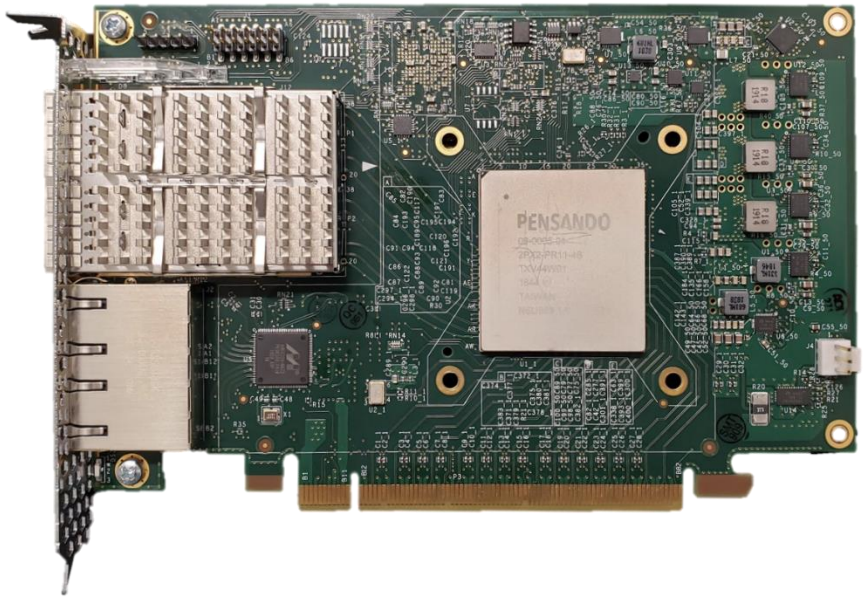


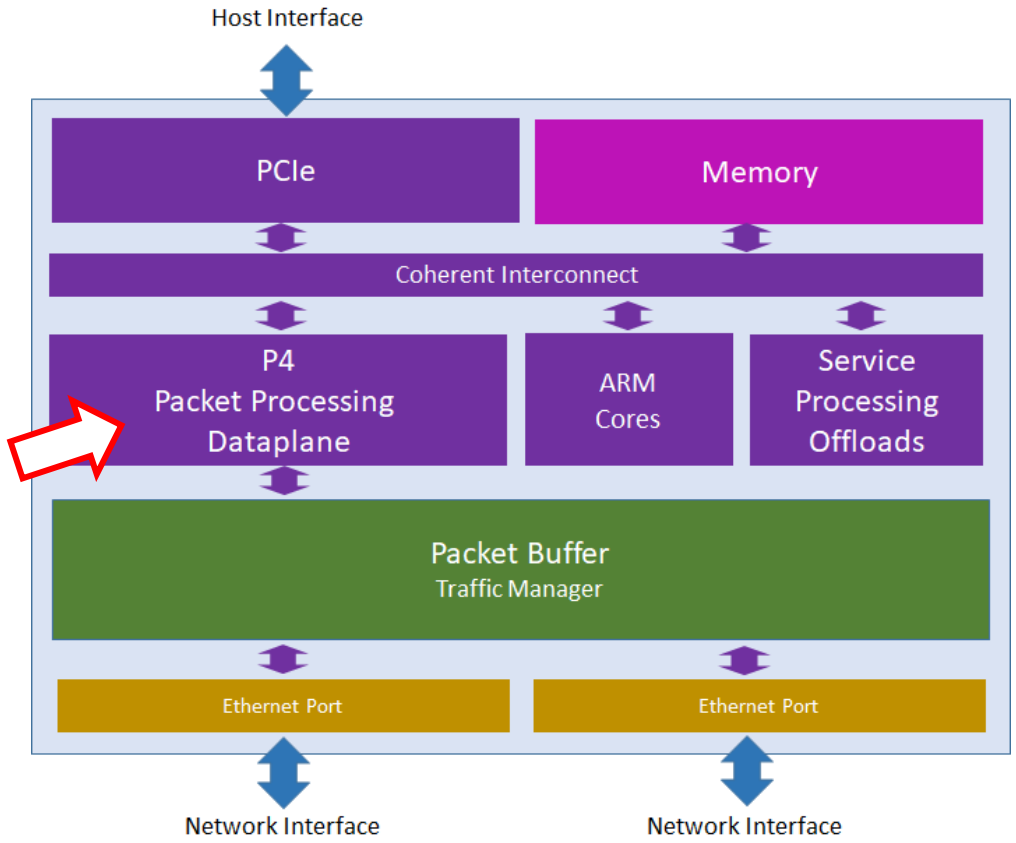
https://www.intel.com/content/www/us/en/programmable/products/boards_and_kits/dev-kits/altera/intel-fpga-pac-n3000/overview.html

Pensando ... P4 + ARM on SmartNIC

P4 + ARM Core による分散処理への P4 の活用

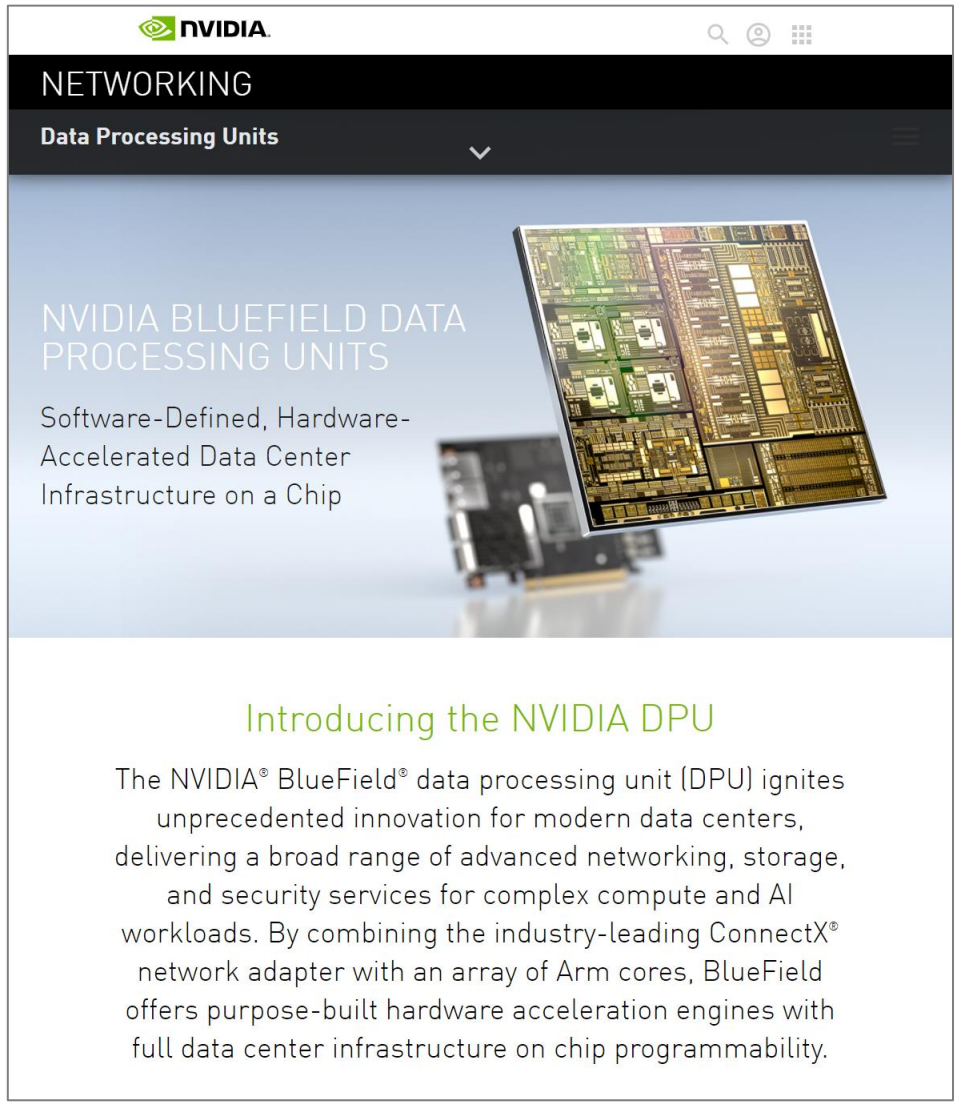
Distributed Services Card (DSC)



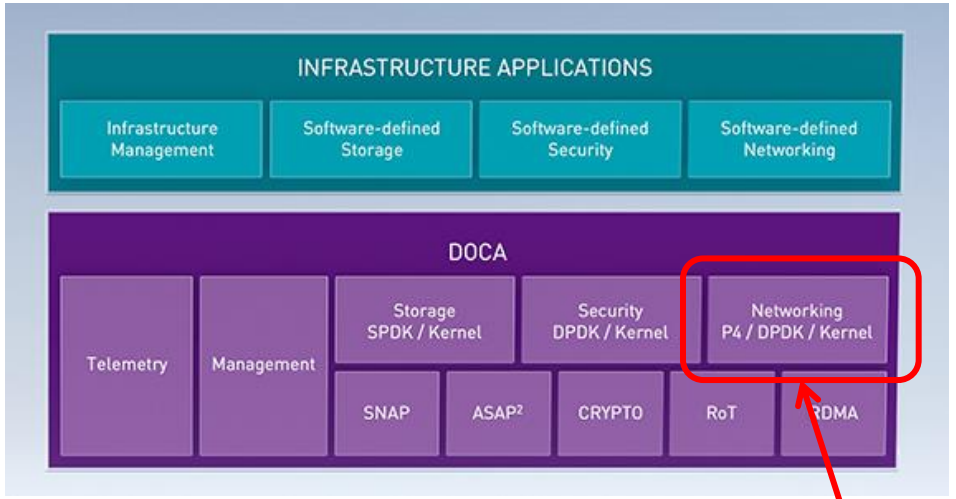


https://p4.org/p4/pensando-joins-p4.html

NVIDIA (Mellanox) BlueField-2 DPU



https://www.nvidia.com/en-us/networking/products/data-processing-unit/



DOCA Software Development Kit

- Network acceleration SDK: ASAP² SDN, emulated VirtIO, 5T PTP technology
- Security acceleration SDK: Inline cryptography, deep packet inspection
- Storage acceleration SDK: Storage emulation and virtualization, crypto, and compression
- RDMA acceleration SDK: UCX, RDMA verbs, GPUDirect
- I/O programmability SDK: P4 and P4RT, OpenSNAPI, cuNE

P4 Support !?
（SmartNIC）

https://www.nvidia.com/content/dam/en-zz/Solutions/Data-Center/documents/datasheet-nvidia-bluefield-2-dpu.pdf

p4.org（本家）Updates

- 運営体制の変更
  - Technical Steering Team (TST) の交代
  - P4 Advisory Board の新設

- P4言語と関連仕様：https://p4.org/specs/
  - （P4 Language and Related Specifications）
  - P4言語：P4_16 v1.2.0 (2019/10/30), v1.2.1 (2020/07/11)
  - P4Runtime：v1.2 release (2020/07/13)
  - In-Band Network Telemetry (INT)：v2.1 (2020/06/15)
  - Telemetry Report Format：v2.0 (2020/06/15)

https://p4.org/p4/tst-elections.html



Please join me in congratulating the newly elected members of the P4 TST for 2020 - 2021: Andy Fingerhut, Stefan Heule and Robert Soule!

P4 TST 2020 - 2021

P4 Technical Steering Team

Andy Fingerhut (new)
Nate Foster
Stefan Heule (new)
Larry Peterson (ex-officio)
Robert Soule (new)
Amin Vahdat

若返り？によりさらに活性化

P4 Advisory Board（新設）

Nate Foster
Nick McKeown
Guru Parulkar
Jen Rexford
Amin Vahdat

P4_16 v1.2.0 解説BLOG : https://p4.org/p4/p4-16-v1.2.0/

A. Appendix: Revision History

Release	Release Date	Summary of Changes
1.0.0	May 17, 2017	Initial version.
1.1.0	November 26, 2018	Added top-level functions, optional and named parameters, enum representations, parser value sets, type definitions, saturating arithmetic, and structured annotations. Removed globalname annotation and added a table size property. Clarified semantics of operations on invalid headers, added restrictions on arguments to calls, and modified precedence of bitwise operators.
1.2.0	October 14, 2019	Added error ParserInvalidArgument, order of const entries, header size methods, 1-bit signed values, signed bit slices, empty tuples, @deprecated annotation, free-form annotations, int type table.apply().miss, string type.
1.2.1	June 11, 2020	Added structure-value expressions, default values, concatenation, structured annotations; standardized several new annotations; generalized typing rule for masks; restricted typing rule for shifts with infinite-precistion operands; clarified evaluation order for table keys; clarified copy-out behavior; clarified semantics of invalid header stacks; clarified initialization semantics; fixed several small issues in grammar.

A.1. Summary of changes made in version 1.2.1

- Added structure-value expressions (Section 8.12).
- Added support for default values (Section 7.3).
- Added support for concatenating signed strings (Section 8.6.1).
- Added key-value and list-structured annotations (Section 18).
- Added @pure and @noSideEffects annotations (Section 18.3.6).
- Added @noWarn annotation (Section 18.3.8).
- Generalized typing for masks to allow serializable enums (Section 8.13.3).
- Restricted the right operands of bit shifts involving infinite-precision integers to be constant and positive (Section 8.7).
- Clarified copy-out behavior for return (Section 11.4) and exit (Section 11.5) statements.
- Clarified semanti… writing-invalid-he…
- Clarified initializa… variables.
- Clarified evaluatio…
- Fixed grammar to… parser (Section 12…

A.2. Summary of changes made in version 1.2.0

- Added table.apply().miss (Section 13.2.2).
- Added string type (Section 7.1.5).
- Added implicit casts from enum values (Section 8.3).
- Allow 1-bit signed values
- Define the type of bit slices from signed and unsigned values to be unsigned.
- Constrain default label position for switch statements.
- Allow empty tuples.
- Added @deprecated annotation.
- Relaxed the structure of annotation bodies.
- Removed the @pkginfo annotation, which is now defined by the P4Runtime specification.
- Added int type (Section 7.1.6.5).
- Added error ParserInvalidArgument (Sections 12.8.2, 12.8.4).
- Clarified the significance of order of entries in const entries (Section 13.2.1.4).
- …te header size (Section 8.16).

https://p4.org/p4-spec/docs/P4-16-v1.2.1.html#sec-revision-history

P4Runtime v1.2 解説BLOG：https://p4.org/p4/p4runtime-v1.2-release.html

Announcing the P4Runtime v1.2 release

Posted by Antonin Bas on July 13, 2020

We are excited to announce the release of P4Runtime v1.2.0. We have come a long way since the creation of the P4 API Working Group in 2017 and the P4Runtime 1.0.0 release in 2019! Congrats to everyone in the P4 API Working Group for making this possible and continuing to improve the quality of the P4Runtime API and specification document.

The end of 2019 was big for P4Runtime, with ONF announcing that the Stratum project was released as open-source. This makes Stratum the first open switch OS to offer P4Runtime as a northbound interface, with multiple hardware platforms already supported. If you do not have access to hardware, fret not as Stratum also supports the bmv2 simulator, which anyone can run on a Linux system.

The first half of 2020 was also very exciting, with 2 minor releases for P4Runtime, v1.1 (Feb 2020) and v1.2 (July 2020).

Notable Changes in P4Runtime v1.1 and v1.2

For the full list of changes, refer to the revision history.

Major Overhaul of Master-Arbitration (v1.1)

The P4Runtime interface allows multiple controllers to be connected to the P4Runtime server running on the device at the same time: one master controller - the only one with write access to the device - and potentially several stand-by controllers. Each controller has a numerical election ID, usually assigned through a leader election algorithm (e.g. Paxos, as implemented in Chubby), which it advertises to the server, with the master being the controller with the highest election ID. In case the master goes offline, one of the stand-by controllers is available to take its place. In P4Runtime v1.0, this hand-over was automatic and handled entirely by the server: upon detecting that the master had disconnected, the server would look for the controller with the next highest election ID, and appoint it in its place. The P4 API Working Group received feedback from Google and ONF engineers implementing SDN controllers that this could lead to issues in production deployments. In particular, in case of a network partition

A. Appendix

A.1. Revision History

A.1.1. Changes in v1.2.0

- Add new `OPTIONAL` match kind. At the moment, `OPTIONAL` is only supported by the v1model architecture [38], and not by PSA. It will eventually be included in the core P4 language.
- Add support in P4Info for structured annotations, which are used to annotate objects with key-value lists or expression lists.
- Add a new `metadata` field of type `bytes` to `TableEntry`. This is more flexible than the now deprecated `controller_metadata` field.
- Add the ability to change the ID of table match fields, action parameters, Packet IO metadata fields, and Value Set match fields in P4Info by using the `@id` annotation.
- Clarify the behavior of some corner cases involving action profiles and selectors, including the watch port feature.
- Support using `string` as the controller type in the `@p4runtime_translation` annotation. Update syntax when using a fixed-width unsigned bitstring as the controller type.

https://p4.org/p4runtime/spec/v1.2.0/P4Runtime-Spec.html#sec-revision-history

https://github.com/p4lang/p4-applications/blob/master/docs/INT_v2_1.pdf

In-band Network Telemetry (INT) Dataplane Specification

Version 2.1

The P4.org Applications Working Group. Contributions from
Alibaba, Arista, CableLabs, Cisco Systems, Dell, Intel, Marvell, Netronome, VMware

2020-06-15

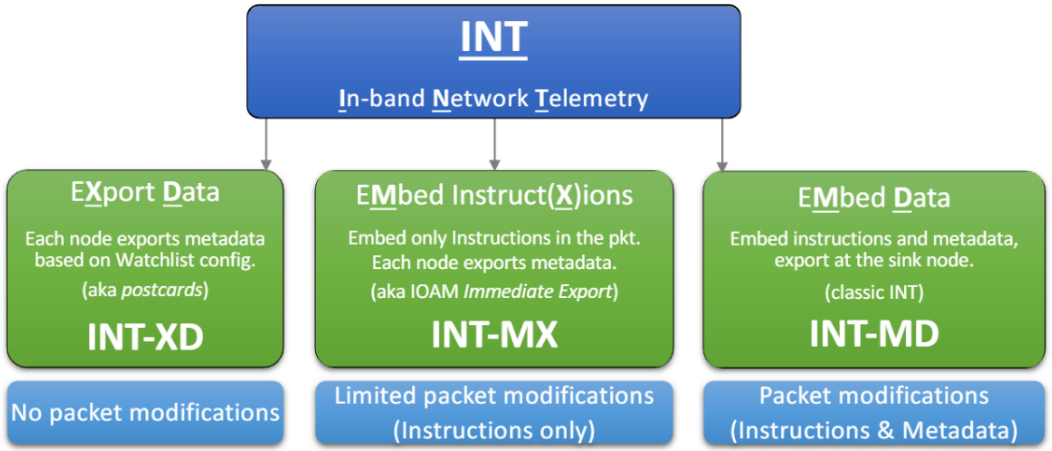


Figure 1. Various modes of INT operation.

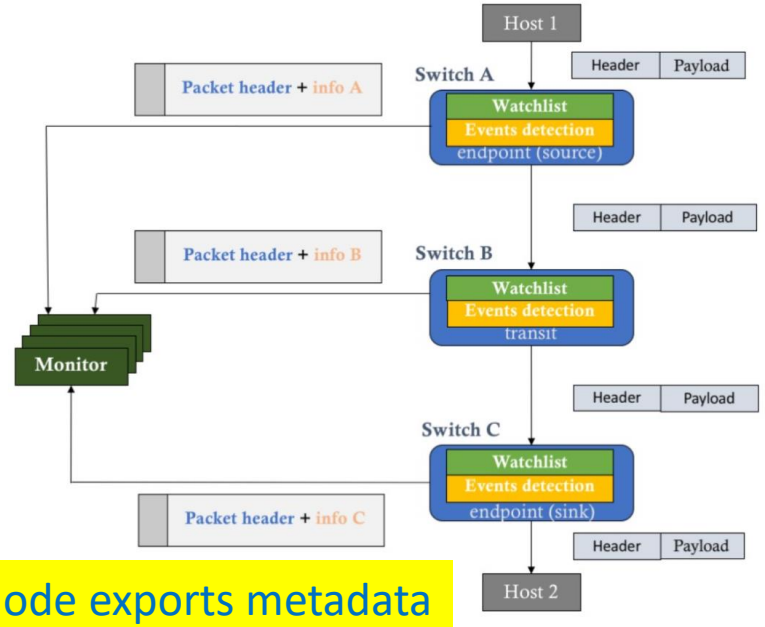
https://github.com/p4lang/p4-applications/blob/master/docs/telemetry_report_v2_0.pdf

Telemetry Report Format Specification

Version 2.0

The P4.org Applications Working Group
Contributions from *CableLabs, Cisco Systems, Intel, VMware, Xilinx*

2020-06-15



INT-XD: Each node exports metadata

Figure 1. INT-XD mode - Telemetry Architecture with per hop reports generated by each node

Telemetry Report Format Specification Version 2.0

https://github.com/p4lang/p4-applications/blob/master/docs/telemetry_report_v2_0.pdf

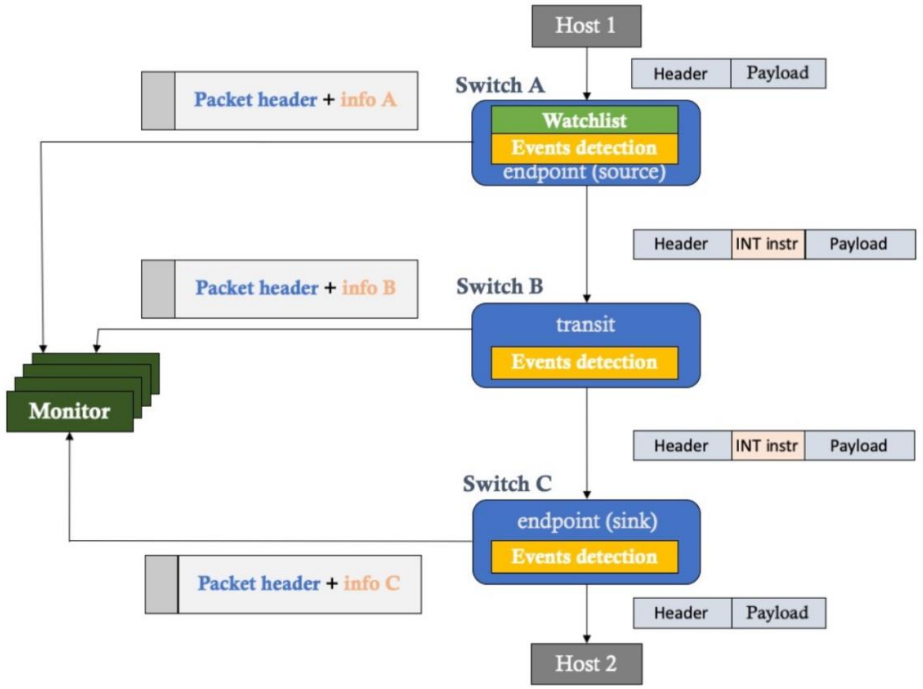


Figure 2. INT-MX mode - Telemetry Architecture with per hop reports generated by each node

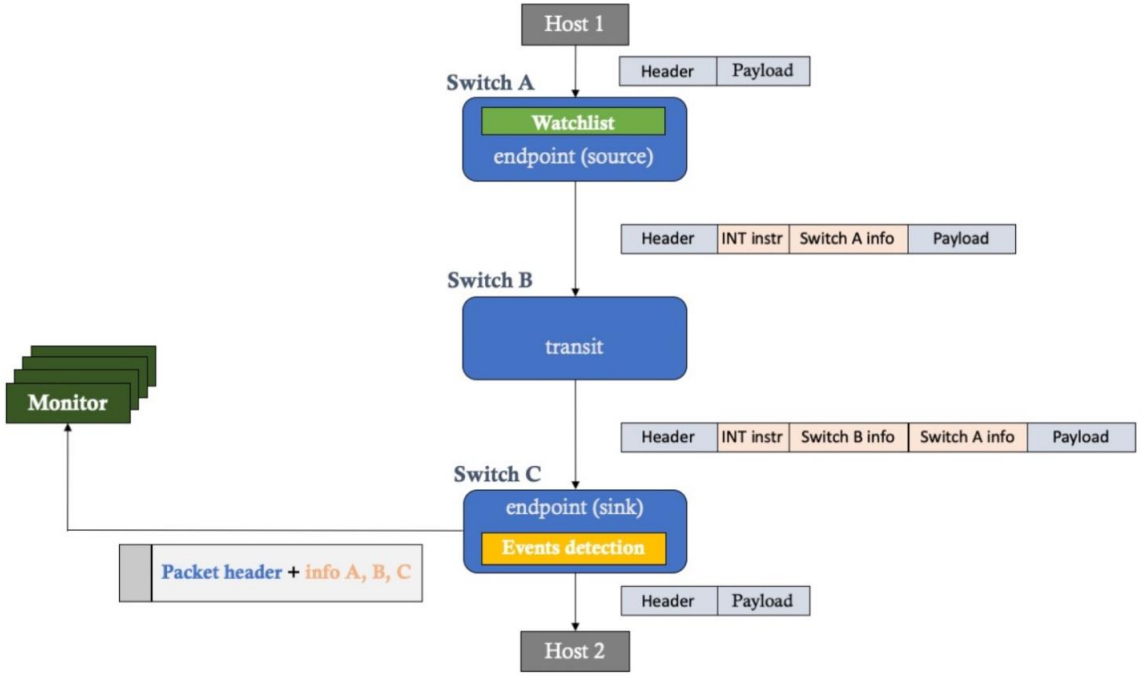


Figure 3. INT-MD mode - Telemetry Architecture with stacked reports generated by sink nodes

INT-MX: Embed only Instructions in the pkt

INT-MD: Embed Instructions & Metadata in the pkt

Kaloom ... 5G UPF に P4 を活用（モバイル）

Kaloom's 5G UPF

The industry's first standalone, highly scalable, lowest latency, multi-tenant solution at 10x TCO reduction.



Automated & Programmable Fabric | Integrated Virtual Router | Ultra-low Latency | 10x TCO Reduction | Native End-to-End Slicing Support | High Capacity UPF with Multi-terabit Throughput

Kaloom's products are based on a clear separation between the control plane and the data plane parts. Our control plane is written in Golang, [illegible] containers over a Kubernetes clustering framework. Our data plane is written in P4. Without a doubt, Kaloom has built the most modern networking solution in the industry. The main advantages resulting from our implementation architecture are modularity, scalability, flexibility, and increased productivity compared to traditional C-Linux networking solutions. Moreover, the utilization of P4-enabled switches results in significant characteristic improvements in terms of throughput and latency. The ability of Kaloom to migrate the data plane components of virtual network functions from x86 toward these Tb/s P4 switches provides substantial CAPEX and OPEX reductions to our customers.

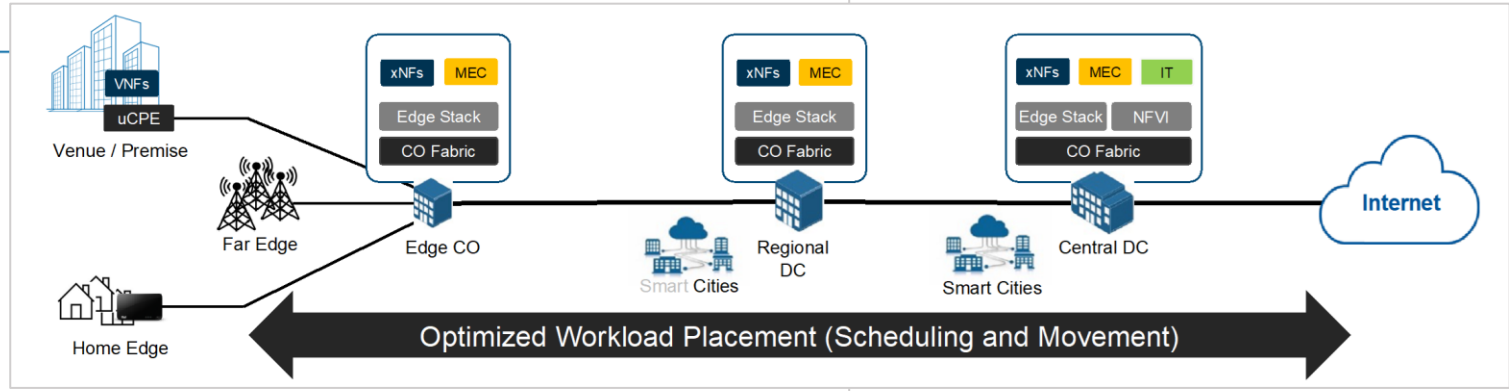
Kaloom has created an edge data center fabric capable of running any L2-L7 network function at-scale with unprecedented throughput and low-latency. By migrating network functions such as UPF, load balancing and others, Kaloom frees-up many servers to run revenue-generating applications rather than internal infrastructure services.

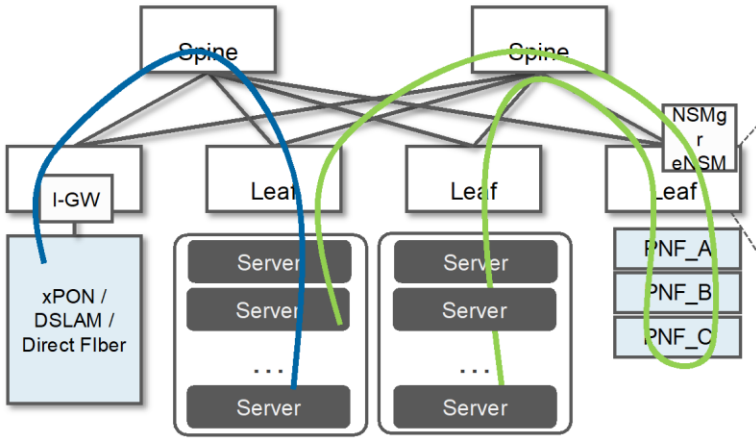
https://www.kaloom.com/blog/kalooms-startup-journey-2

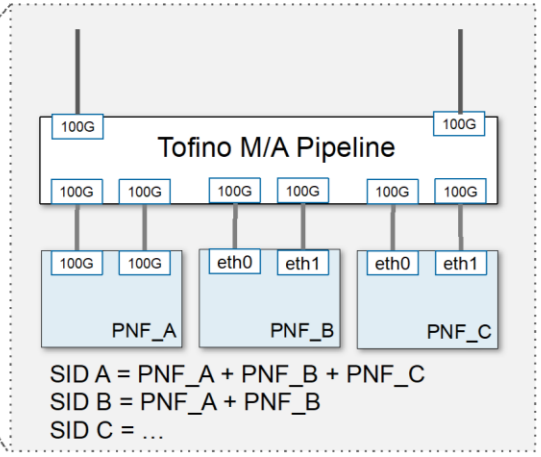
Bell Canada / NoviFlow ... Service Chaining (SRv6) using Tofino (P4)

PNF Integration

- Leverage SRv6 to make PNF pipeline reachable (virtually inline)
- Leverage SRTE to create associated policies within network
- Leverage BGP or NSM to create a Service Registry
- Independent if the PNF is SR aware or not







NoviFlow

Powered by NoviFlow CyberMapper

Bell

https://www.youtube.com/watch?v=035VgMt-9b0

LAYWER123 World Congress 2020

“Service programming with SRv6 - NoviFlow/Bell Canada”

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再掲

「パケット転送」以外のユースケースへの適用

Key-value store

NetCache : Balancing Key-Value Stores with Fast In-Network Caching Key-value stores power online services.

Jin, Xin; Li, Xiaozhou; Zhang, Haoyu; Soule, Robert; Lee, Jeongkeun; Foster, Nate; Kim, Changhoon; Stoica, I. (2017). Sosp, (Figure 1), 121?136. https://doi.org/10.1145/3132747.3132764

Map Reduce

Network Map Reduce.

Song, H., Gong, J., & Chen, H. (2016).
Retrieved from http://arxiv.org/abs/1609.02982

DAIET ... Data Aggregation In nETwork

https://sands.kaust.edu.sa/daiet/
Poster: Amedeo Sapio, Ibrahim Abdelaziz, Marco Canini and Panos Kalnis.
“DAIET: A System for Data Aggregation Inside the Network.”
In ACM Symposium on Cloud Computing 2017, Santa Clara, California, USA.

コンセンサス（合意）

Paxos Made Switch-y.

Dang, H. T., Canini, M., Pedone, F., & Soule, R. (2016).
ACM SIGCOMM Computer Communication Review, 46(1), 18~24. https://doi.org/10.1145/2935634.2935638

Eris: Coordination-Free Consistent Transactions Using In-Network Concurrency Control.

Li, J., Michael, E., K Ports, D. R., & K, D. R. (2017).
Sosp. https://doi.org/10.1145/3132747.3132751

NetPaxos: Consensus at Network Speed.

Dang, H. T., Sciascia, D., Canini, M., Pedone, F., & Soule, R. (2015).
Proceedings of the 1st ACM Sigcomm Symposium on SDN Research (SOSR), 5:1--5:7.
https://doi.org/10.1145/2774993.2774999

NetChain: Scale-Free Sub-RTT Coordination.

Nsdi, 35~49. Retrieved from http://dl.acm.org/citation.cfm?doid=3132747.3132761

様々なTargetに対する P4C Backend の開発

- p4ubpf: p4c-ubpf: a New Back-end for the P4 Compiler June 1, 2020
  - https://p4.org/p4/p4c-ubpf.html

NEW!!

Userland BPF ... can be used in OVS

- p4c-ebpf
  - https://github.com/p4lang/p4c/tree/master/backends/ebpf
- p4c-xdp: (by VMware)
  - https://github.com/vmware/p4c-xdp
  - https://qiita.com/hibitomo/items/3ed846beb2e504f0ffb6
- DPDK (Multi Target P4C)
  - http://p4.elte.hu/
  - https://github.com/P4ELTE/t4p4s

P4関連論文紹介 パケット転送・オフロード（2019～2020）

- Luo, S., Yu, H., Li, K., & Xing, H. (2020). Efficient File Dissemination in Data Center Networks With Priority-Based Adaptive Multicast. *IEEE Journal on Selected Areas in Communications*, *38*(6), 1161–1175. https://doi.org/10.1109/JSAC.2020.2986616
- Zhang, K. (2020). Gallium : Automated Software Middlebox Offloading to Programmable Switches. *Sigcomm'20*, 283–295.
- Kim, D., Kim, C., & Lee, J. (2020). *TEA : Enabling State-Intensive Network Functions on Programmable Switches*. 90–106. https://dl.acm.org/doi/pdf/10.1145/3387514.3405855
- Zhou, D. (2019). Data Structure Engineering for High Performance Software Packet Processing. *CMU PHD Thesis*. https://www.cs.cmu.edu/~dongz/proposal.pdf
- Zhang, C., Bi, J., Zhou, Y., & Wu, J. (2019). HyperVDP: High-Performance Virtualization of the Programmable Data Plane. *IEEE Journal on Selected Areas in Communications*, *37*(3), 556–569. https://doi.org/10.1109/JSAC.2019.2894308
- Kundel, R., Nobach, L., Blendin, J., Kolbe, H.-J., Schyguda, G., Gurevich, V., Koldehofe, B., & Steinmetz, R. (2019). P4-BNG: Central Office Network Functions on Programmable Packet Pipelines. *2019 15th International Conference on Network and Service Management (CNSM)*, 1–9. https://doi.org/10.23919/CNSM46954.2019.9012666
- Tokusashi, Y., Dang, H. T., Pedone, F., Soul, R., & Zilberman, N. (2019). The case for in-network computing on demand. *Proceedings of the 14th EuroSys Conference 2019*. https://doi.org/10.1145/3302424.3303979

Zhang, K. (2020). Gallium : Automated Software Middlebox Offloading to Programmable Switches. Sigcomm'20, 283–295

C++ => P4 & C++ 変換フロー

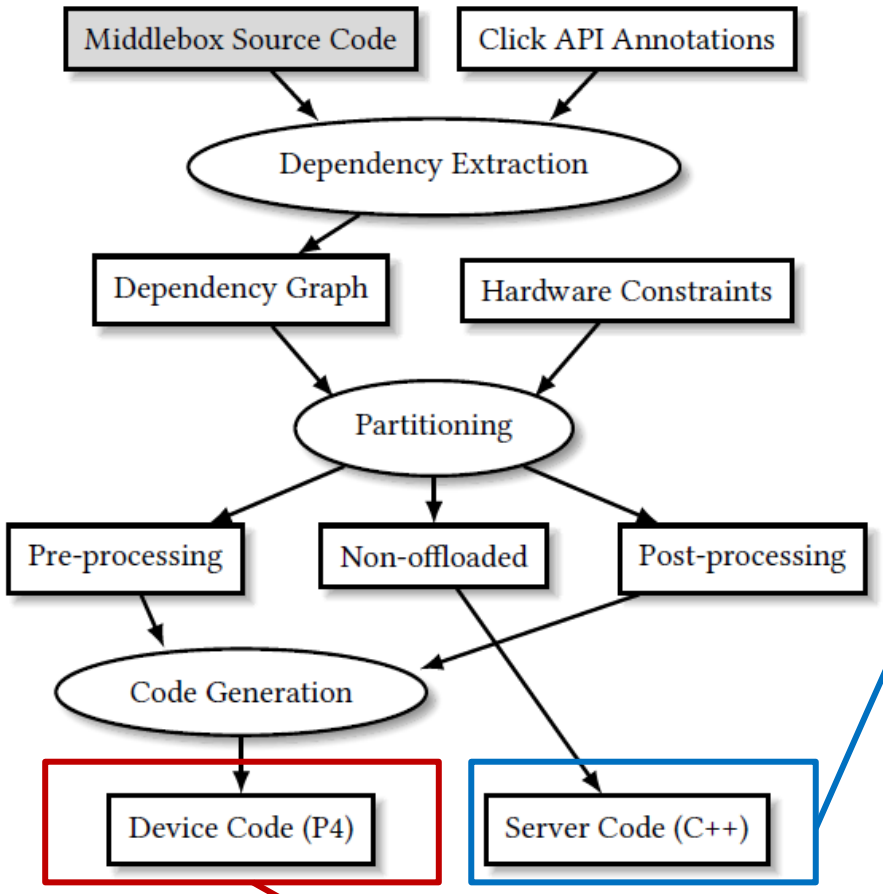


Figure 2: Workflow of Gallium. The shaded box is the input program written by middlebox programmers. Other rectangular boxes are annotation and configuration inputs given to Gallium, intermediate representations, and final outputs.

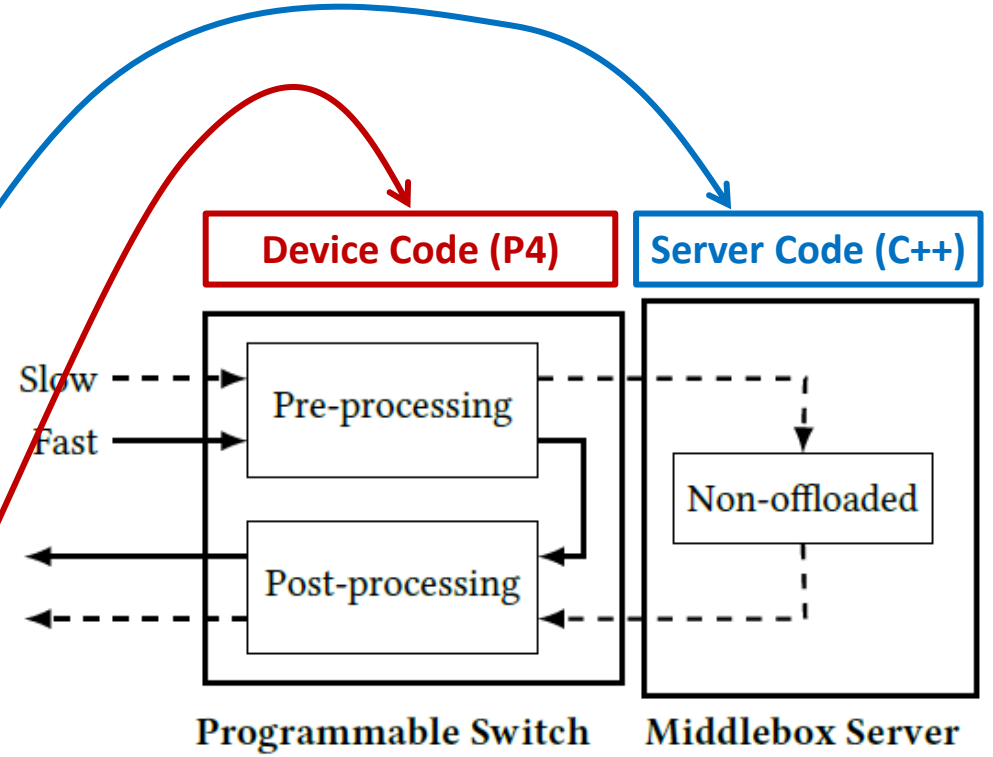


Figure 1: Packet flow in an offloaded middlebox.

実行時のコード配置

C++ Mifflebox Source Code をClick Router API Annotationを元にP4 Targetで動作可能な機能をオフロードするP4 Code を生成

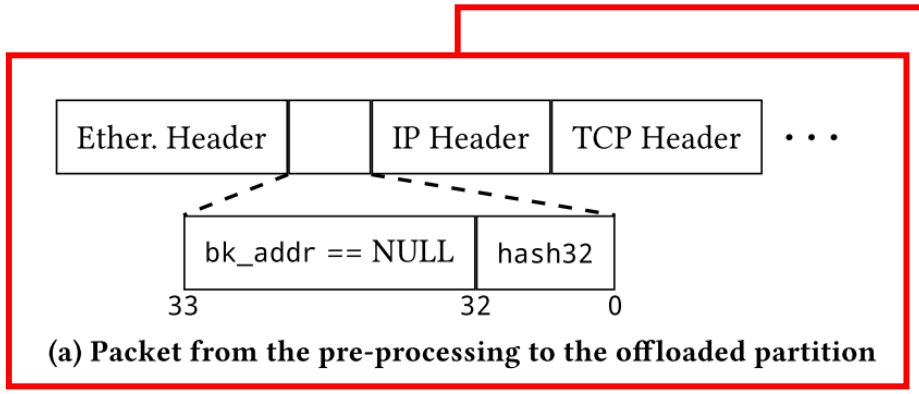


Figure 5: The packet format used between programmable switch and middlebox server in `MiniLB`.

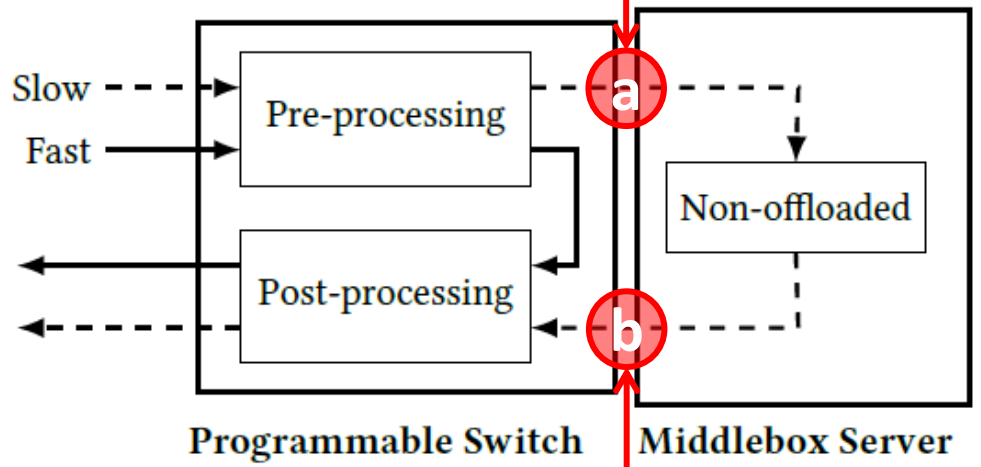


Figure 1: Packet flow in an offloaded middlebox.

Zhang, K. (2020). Gallium : Automated Software Middlebox Offloading to Programmable Switches. Sigcomm'20, 283–295

Source Code 削減割合

Middlebox	Input (C++)	Output (P4)	Output (C++)	Output (Total)	Code Reduction	
MazuNAT	1687	516	579	1095	35%	MazuNAT
Load Balancer	1447	522	602	1124	28%	Load Balancer
Firewall	1151	506	403	909	21%	Firewall
Proxy	953	292	279	571	40%	Proxy
Trojan Detector	882	571	418	989	-12%	Trojan Detector

This part added by Kentaro Ebisawa

Table 1: Comparison of lines of code for Click-based middleboxes before and after Gallium compiles them.

P4関連論文紹介 P4を活用したアプリケーション（2019〜2020）

In-Network Computing / ML / AI / Deep Learning（ネットワーク内でのデータ処理）

- Jepsen, T., Alvarez, D., Foster, N., Kim, C., Lee, J., Moshref, M., & Soule, R. (2019). Fast String Searching on PISA. *SOSR 2019 - Proceedings of the 2019 ACM Symposium on SDN Research*, 21–28. https://doi.org/10.1145/3314148.3314356
- Chen, J., & Ran, X. (2019). Deep Learning With Edge Computing: A Review. *Proceedings of the IEEE*, *107*(8), 1655–1674. https://doi.org/10.1109/JPROC.2019.2921977
- Lin, L., Liao, X., Jin, H., & Li, P. (2019). Computation Offloading Toward Edge Computing. *Proceedings of the IEEE*, *107*(8), 1584–1607. https://doi.org/10.1109/JPROC.2019.2922285
- Sapio, A., Canini, M., Ho, C.-Y., Nelson, J., Kalnis, P., Kim, C., Krishnamurthy, A., Moshref, M., Ports, D. R. K., & Richtárik, P. (2019). *Scaling Distributed Machine Learning with In-Network Aggregation*. http://arxiv.org/abs/1903.06701
- Xiong, Z., & Zilberman, N. (2019). *Do Switches Dream of Machine Learning?* 25–33. https://doi.org/10.1145/3365609.3365864
- Zhu, H., Bai, Z., Li, J., Michael, E., Ports, D. R. K., Stoica, I., & Jin, X. (2019). Harmonia: Near-Linear Scalability for Replicated Storage with in-Network Conflict Detection. *Proc. VLDB Endow.*, *13*(3), 376–389. https://doi.org/10.14778/3368289.3368301

Fast String Searching on PISA

- CPUで検索を実行する場合 “PCIe” 経由 ⇒ 帯域は狭く遅延は大きい（*1）
  - （*1）PCIe 4.0 x8 => 128Gbps
- ASICで検索を実行する場合 “SERDES” 経由 ⇒ 広帯域・低遅延 （*2）
  - （*2）Tofino ASIC => 100Gbps x32/65 ports

- パケットペイロードの状態で文字列（String）を検索する
  - キーワードを “決定性有限オートマトン（DFA）” 表現に変換
  - DFA を Match/Action Table (TCAM) を用いて実装
  - パケットペイロードの先頭（x bytes）を Match/Action Table で検索
  - 検索済み領域を省いたパケットを再循環（recirculate）して検索
  - （ペイロードなくなるまで継続）

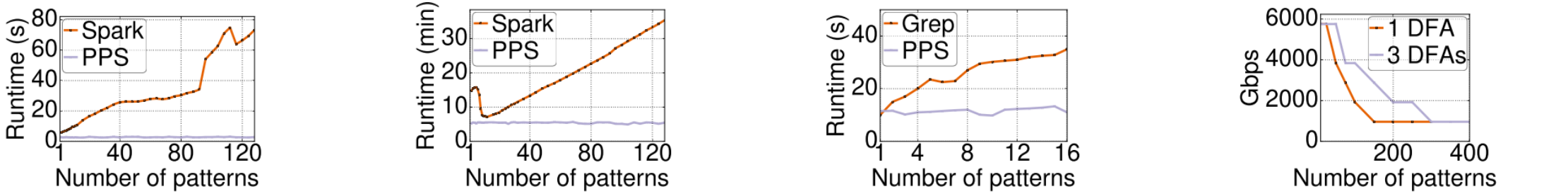
Fast String Searching on PISA



Figure 2: DFAs with different strides for "dog".

Match		Action
state	chars	
0	do	set_state(1)
3	og	accept(4)
1	g*	accept(2)
0	*d	set_state(3)

Figure 3: Table for 2-stride (k=2) DFA



(a) Filtering e-mails. **(b) Filtering tweets.** **(c) Searching a 12GB log file.** **(d) Calculated tput feasibility**

Figure 4: PPS end-to-end experiments and micro-benchmarks.

Jepsen, T., Alvarez, D., Foster, N., Kim, C., Lee, J., Moshref, M., & Soule, R. (2019). **Fast String Searching on PISA.** *SOSR 2019 - Proceedings of the 2019 ACM Symposium on SDN Research*, 21–28. https://doi.org/10.1145/3314148.3314356

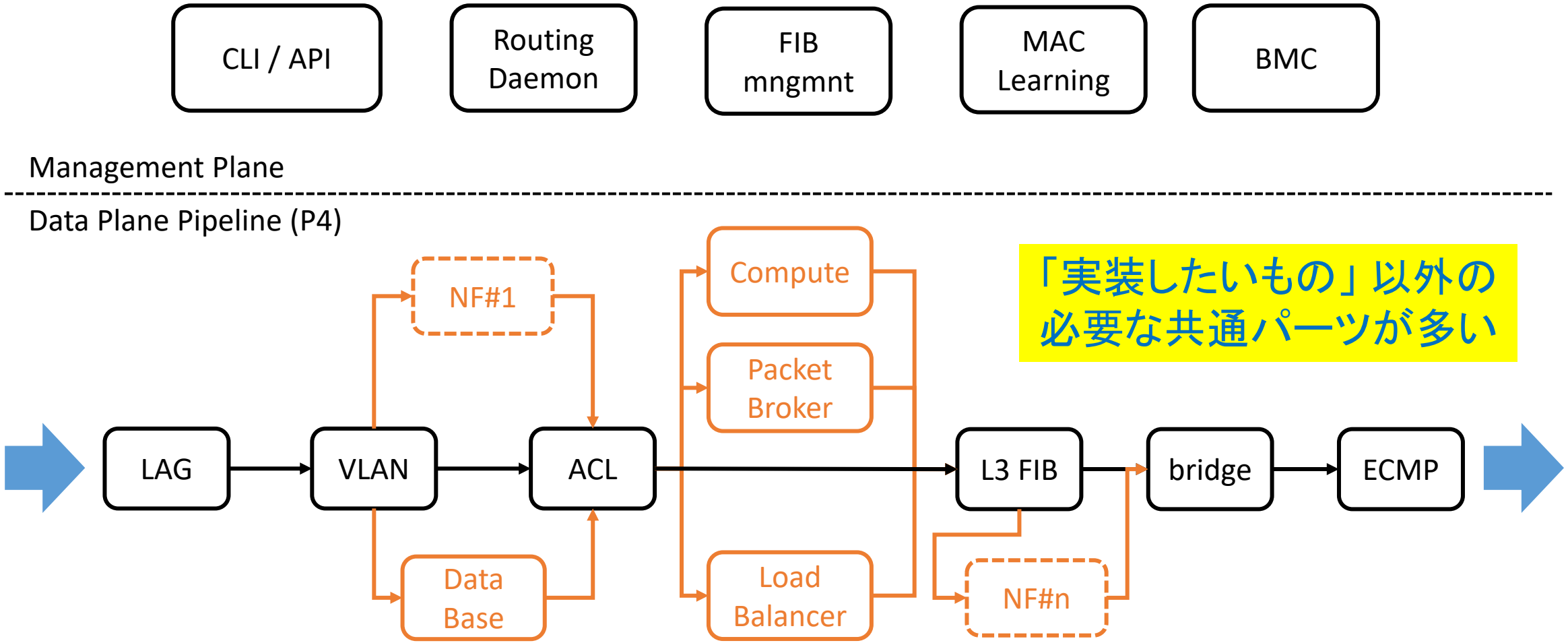
ホワイトボックス・スイッチが広げる「ネットワークの可能性」

データプレーン プログラミング「P4の動向」

「In-Network Computing の研究動向」

ホワイトボックス・スイッチ「今後への期待」

Network Function / Computation の実装



Intel/Barefoot Tofino ASIC 向け P4 プログラムの公開

[P4-announce] Publishing Code Developed for Intel(R) Tofino
Barefoot FASTER research at barefootnetworks.com
Fri May 22 12:50:06 EDT 2020

2020年5月22日

Dear P4 Community:

I am excited to share some incredible news!

What is the announcement?

Intel has given permission to the P4 community to openly publish the data and control plane code written for Tofino.

Who is this for?

This is targeted towards all Tofino users, including Academia and Research organizations that are using Tofino to conduct networking research.

What can you do with this?

You can publish your **P4** or **control plane code** developed for Tofino.

... 略 ...

http://lists.p4.org/pipermail/p4-announce_lists.p4.org/2020/000252.html
http://lists.p4.org/pipermail/p4-discuss_lists.p4.org/2020-May/000544.html

Q: What are Intel's longer-term plans and vision for being more open about Tofino?

A: Intel intends to open Tofino in the long term. As it is the case with general purpose CPUs, we believe that people should be able to learn about the architecture and interfaces, teach about them and write the code openly. This announcement is just the first step. Stay tuned!

- 高速プログラマブル・イーサネット ASIC（Tofino）用プログラムのオープンソース公開・開発が可能に
- 将来的には Intel CPU のように内部情報も公開！？

※ 参考：Intel SDM (Software Developer Manuals)
https://software.intel.com/content/www/us/en/develop/articles/intel-sdm.html

RARE ... TNA based Open Source Router

https://wiki.geant.org/display/RARE/Home

RARE
Router for Academia Research & Education

"Yet another routing/forwarding network stack ... made by routing/forwarding network engineers ..."

Overview

RARE (**R**outer for **A**cademia, **R**esearch & **E**ducation) is an ongoing effort under the GÉANT 3rd programme which focus on determining if a routing software platform solution can fit **R&E** use cases. The project aims to integrate different pieces of software related to these building blocks:

- **control plane**: **RARE** uses **FreeRouter** under the hood used as the control plane component
- **data plane**: (not only P4) **P4** behavioural language is used to describe the packet processing behaviour of **RARE** data plane
- and **communication interface** between the control plane and data plane: Interface compliant to P4Runtime specification ensure this function

A key part of the work consists in enabling a control plane software to pilot a data plane via a programmatic interface.

Resulting work

- Software interface that maps the forwarding state used by a an Open source control plane (FreeRouter)
- to programmable data plane running RARE software

Tofino Native Architecture (tna.p4)
ベースのオープンソース実装

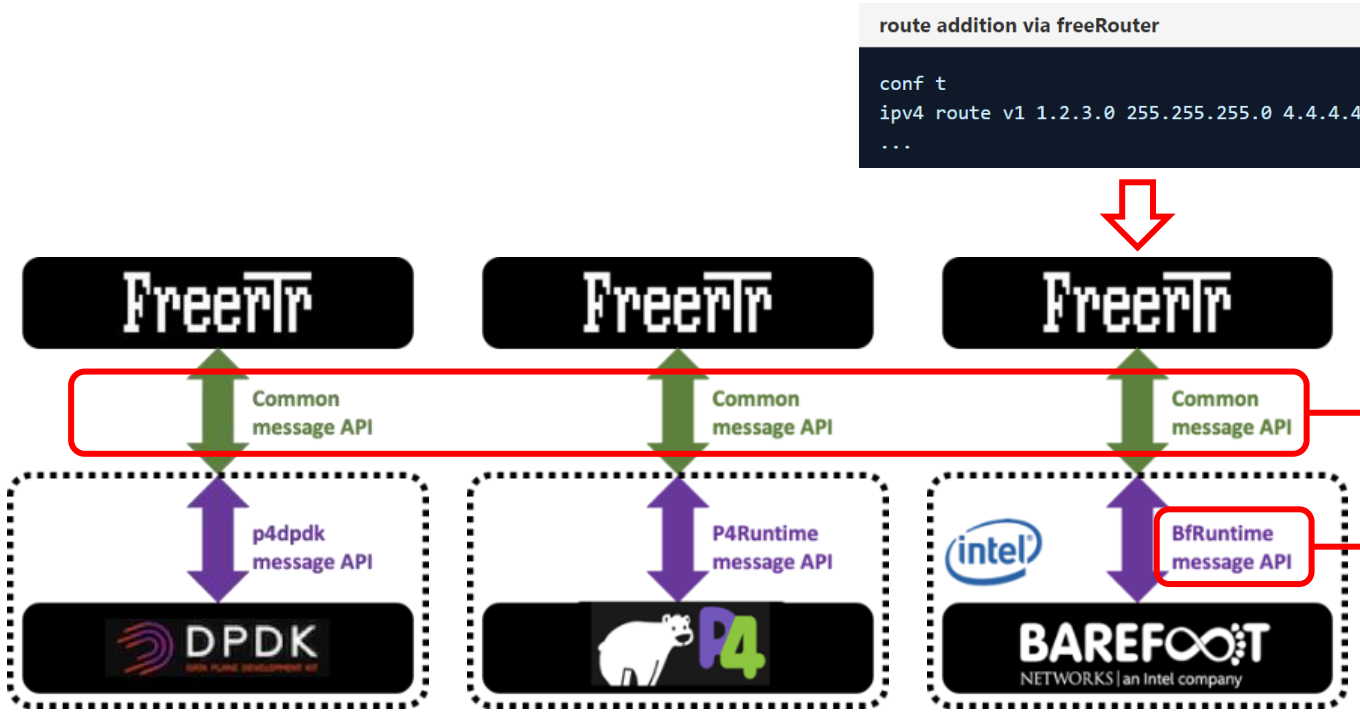
L2 bridge, L3 routing, VLAN, ACL, MPLS, EVPN MPLS, VXLAN, SR-TE and more...

Complete feature list

Type	Test #	Name	BMv2	TNA	DPDK
acl	01	copp	✓	✓	✓
acl	02	ingress access list	✓	✓	✓
acl	03	egress access list	✓	✓	✓
acl	04	nat	✓	✓	✓
acl	05	vlan ingress access list	✓	✓	✓
acl	06	vlan egress access list	✓	✓	✓
acl	07	bundle ingress access list	✓	✓	✓
acl	08	bundle egress access list	✓	✓	✓
acl	09	bundle vlan ingress access list	✓	✓	✓
acl	10	bundle vlan egress access list	✓	✓	✓
acl	11	bridge ingress access list	✓	✓	✓
acl	12	bridge egress access list	✓	✓	✓
acl	13	vlan bridge ingress access list	✓	✓	✓
acl	14	vlan bridge egress access list	✓	✓	✓
acl	15	ingress pppoe access list	✓	✓	✓
acl	16	egress pppoe access list	✓	✓	✓
acl	17	ingress vlan pppoe access list	✓	✓	✓
acl	18	egress vlan pppoe access list	✓	✓	✓
acl	19	hairpin ingress access list	✓	✓	✓

RARE ... TNA based Open Source Router

https://wiki.geant.org/pages/viewpage.action?pageId=148093279



route addition via freeRouter

```
conf t
ipv4 route v1 1.2.3.0 255.255.255.0 4.4.4.4
...
```

message API: route4_add

```
...
rx: ['route4_add', '1.2.3.0/24', '13063', '4.4.4.4', '1', '\n']
...
```

```
if splt[0] == "route4_add":
    addr = splt[1].split("/")
    self.writeForwardRules4(
        1, addr[0], int(addr[1]), int(splt[2]), int(splt[4])
    )
```

```
def writeForwardRules4(self, op_type, dst_ip_addr, dst_net_mask, port, vrf):
    # for any reason, control plane is sending a msg
    # with port=-1
    if port < 0:
        return

    tbl_global_path = "ig_ctl.ig_ctl_ipv4"
    tbl_name = "%s.tbl_ipv4_fib_lpm" % (tbl_global_path)
    tbl_action_name = "%s.act_ipv4_set_nexthop" % (tbl_global_path)

    key_fields = [
        gc.KeyTuple("hdr.ipv4.dst_addr", dst_ip_addr, prefix_len=dst_net_mask),
        gc.KeyTuple("ig_md.vrf", vrf),
    ]
    data_fields = [gc.DataTuple("nexthop_id", port)]
    key_annotation_fields = {"hdr.ipv4.dst_addr": "ipv4"}
    data_annotation_fields = {}
    self._processEntryFromControlPlane(
        op_type,
        tbl_name,
        key_fields,
        data_fields,
        tbl_action_name,
        key_annotation_fields,
        data_annotation_fields,
    )
```

抽象化レイヤーにより
複数のデータプレーンをサポート

Transparent Security @Cable Labs

INT & Closed Loop Automation を組み合わせた異常検知＆対応

Transparent Security https://github.com/cablelabs/transparent-security/

Transparent Security is a solution for identify the source devices of a DDoS attack and mitigates the attack in the customer premises or the access network. This solution leverages a P4 based programmable data plane for add in-band network telemetry (INT) for device identification and in-band mitigation.

TNA based code:

https://github.com/cablelabs/transparent-security/blob/master/p4/aggregate/aggregate_tna.p4
https://github.com/cablelabs/transparent-security/blob/master/p4/core/core_tna.p4

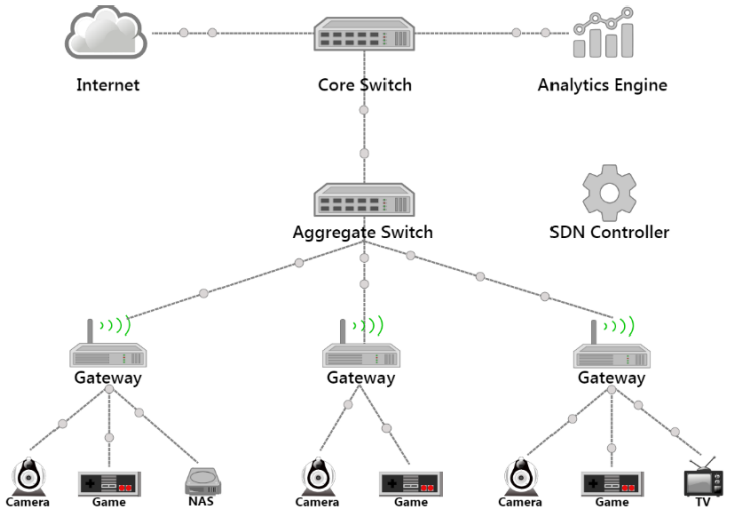
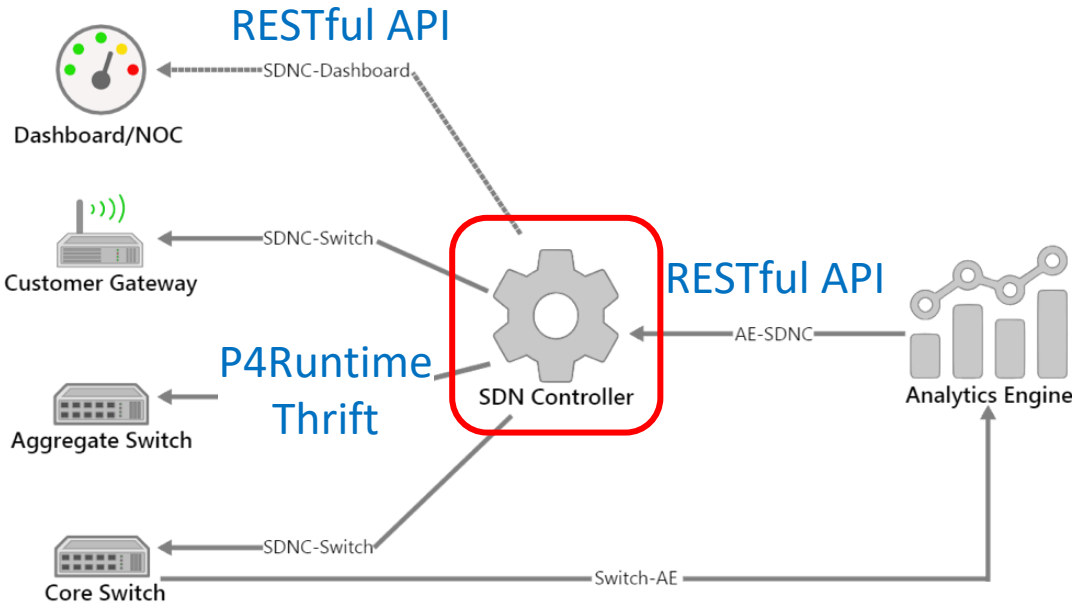


Figure 3 - Architecture Used in the Initial Demonstration of the Transparent Security Concept



Packet with INT header

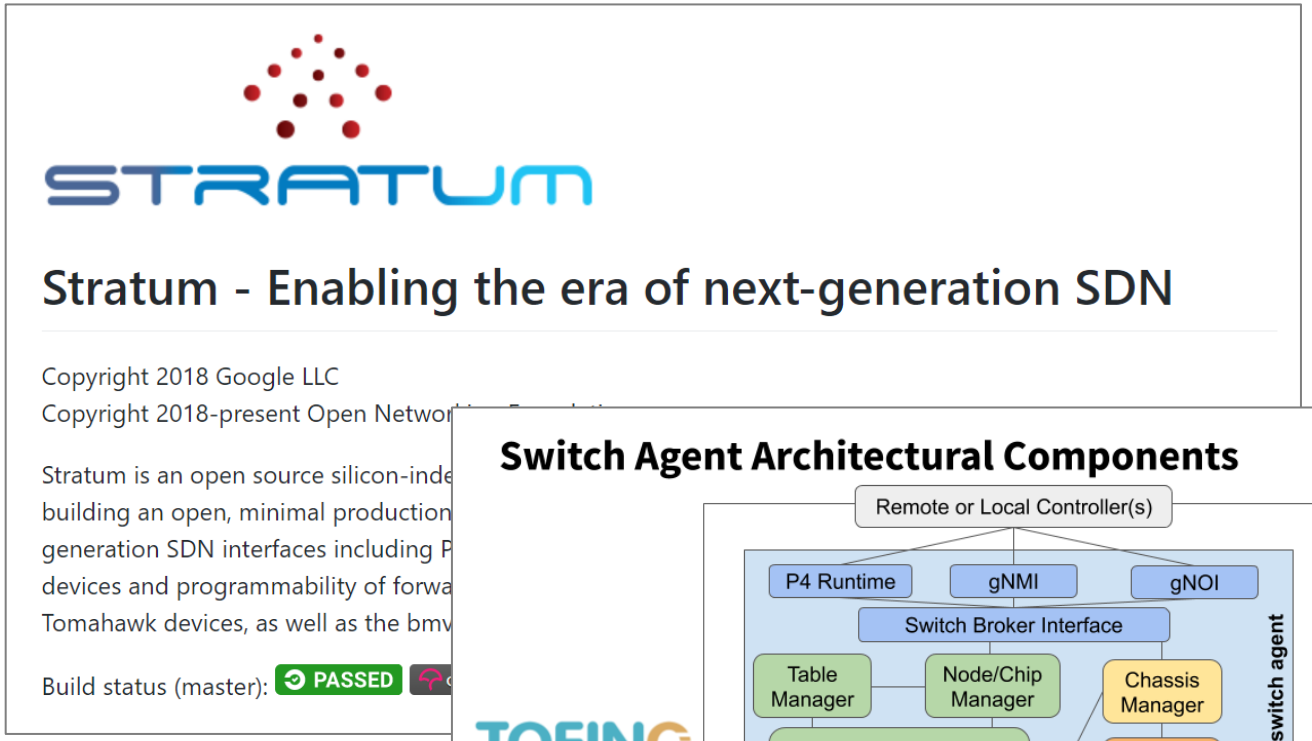
https://www.cablelabs.com/vaccinate-your-network-to-prevent-the-spread-of-ddos-attacks

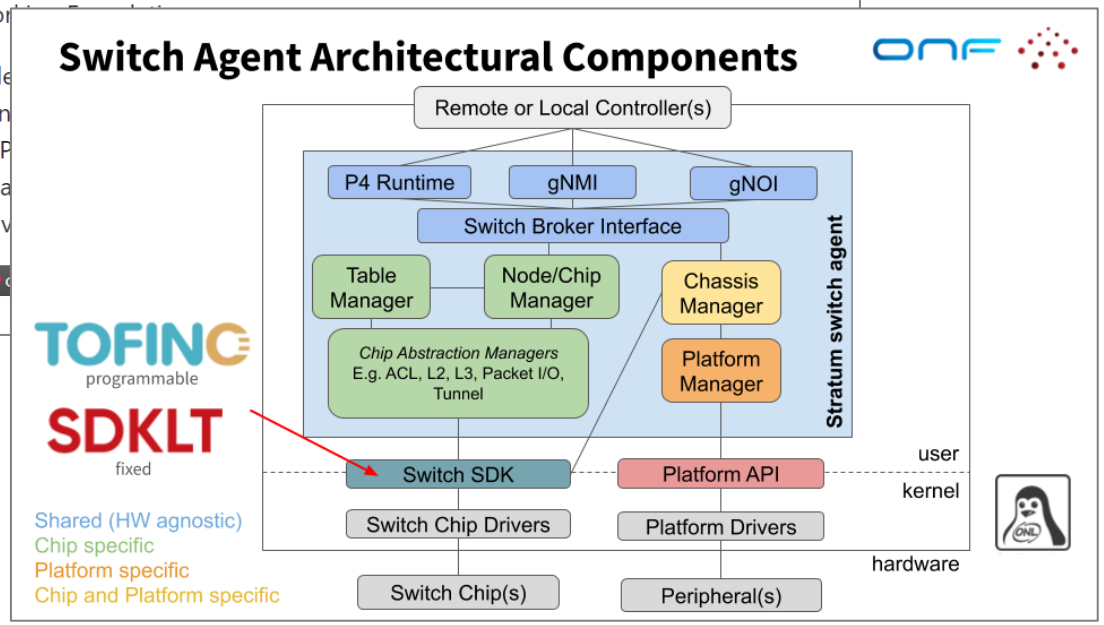
Transparent Security Architecture

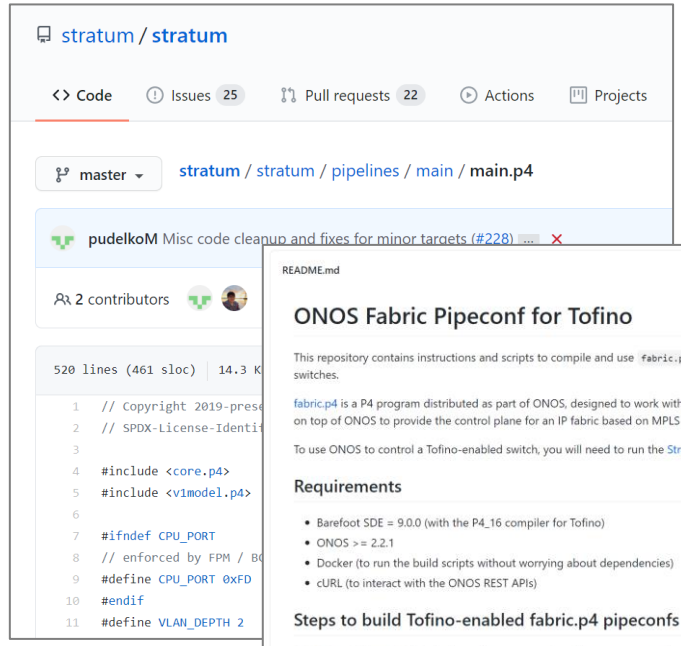
TNA & v1model 両方に対応
抽象化はSDN Controllerを利用

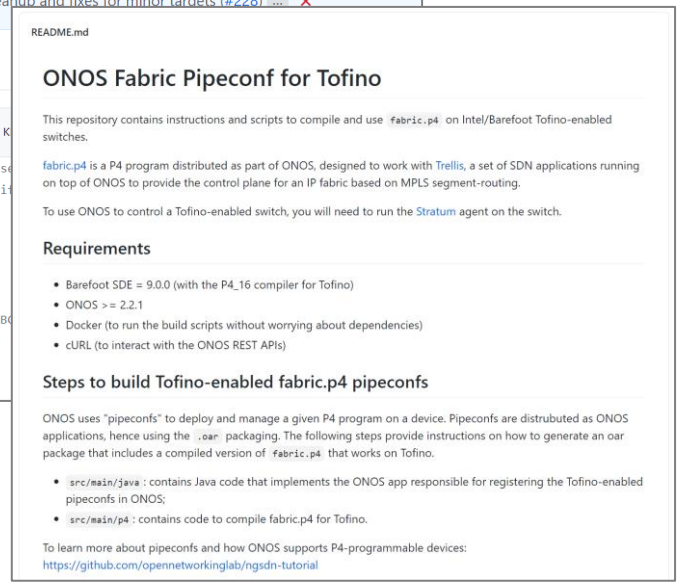
Stratum ... Open Source SDN switch agent

https://github.com/stratum/stratum







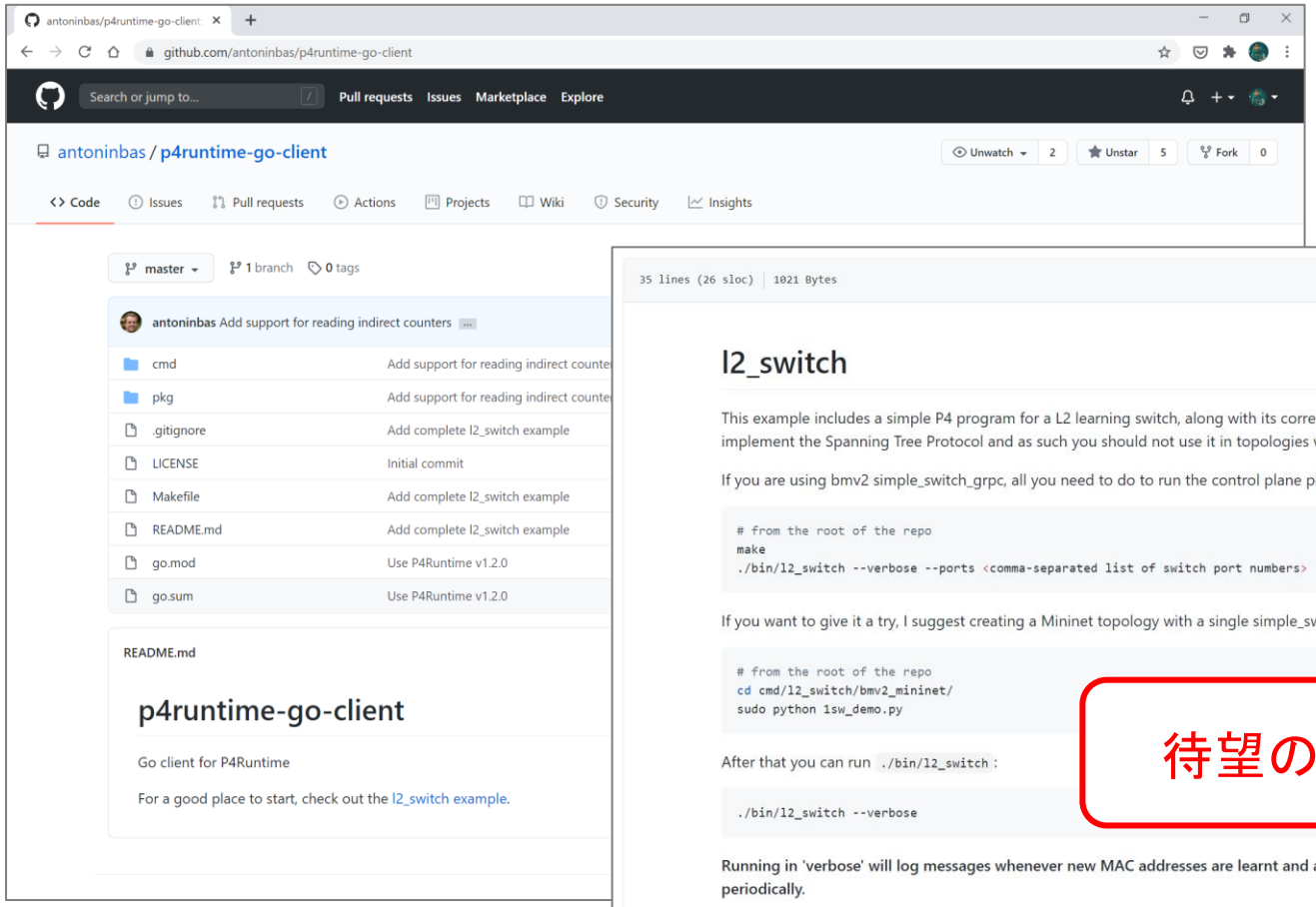


https://github.com/opencord/fabric-tofino

近日 TNA ベースのコードを
リリース予定（という噂）

P4Runtime Go Client example

https://github.com/antoninbas/p4runtime-go-client

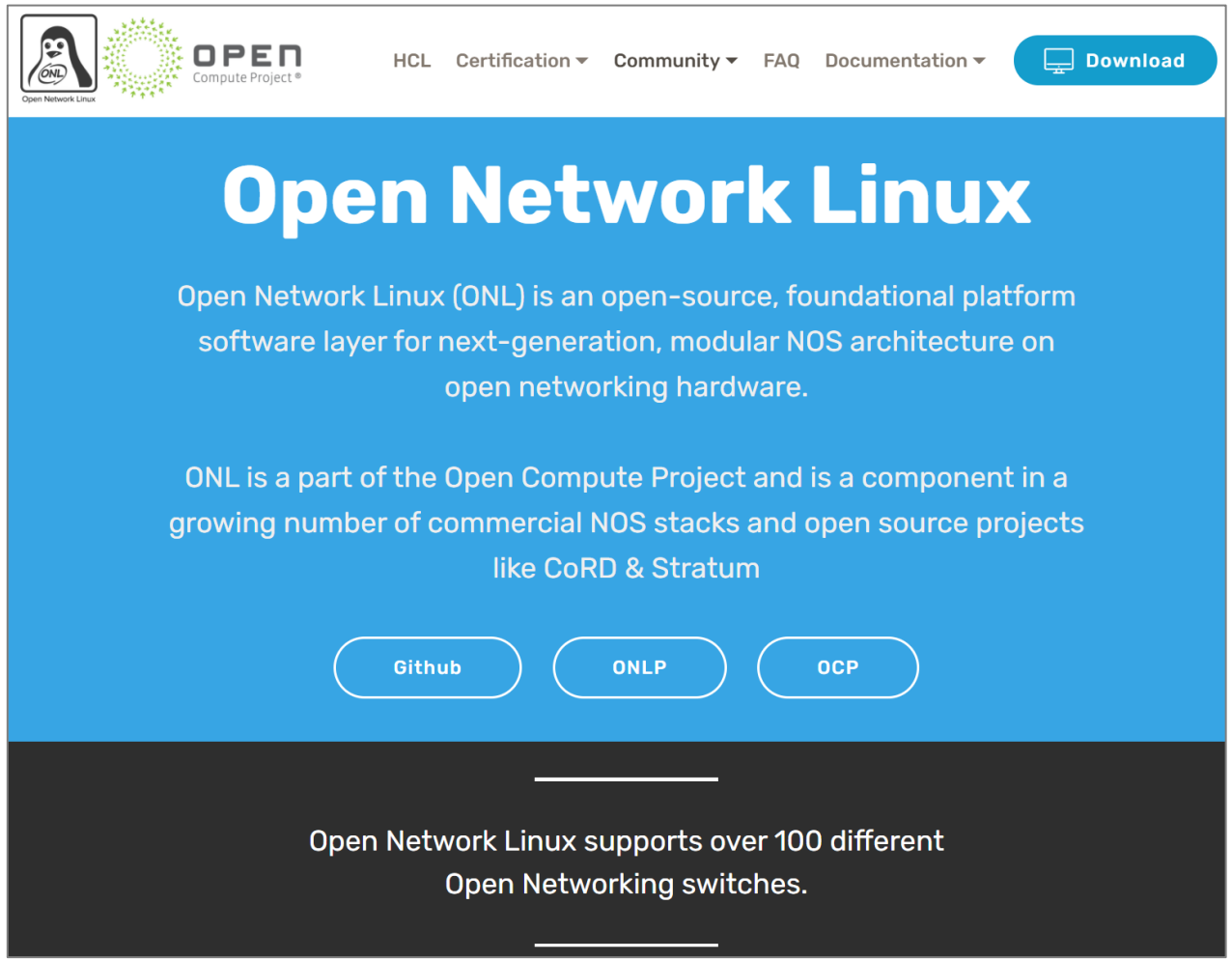






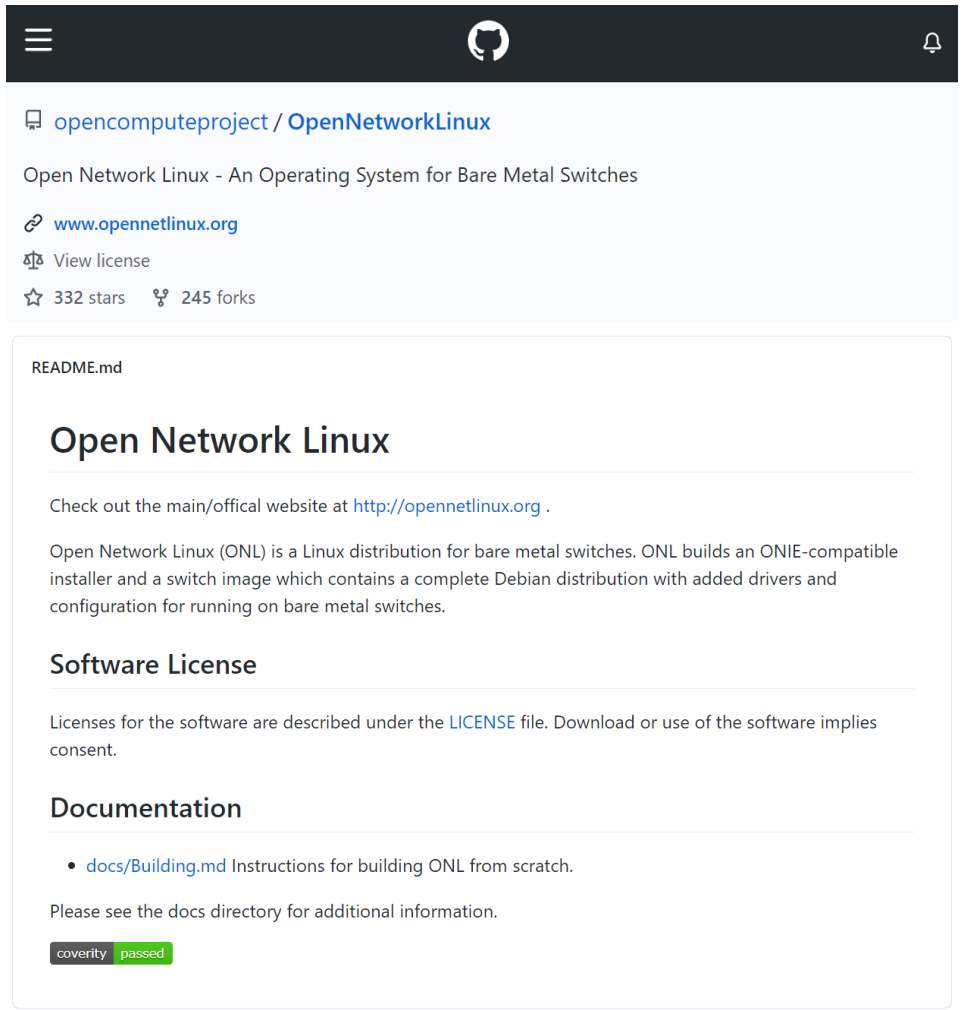
待望の？ GolangによるP4Runtimeサンプル実装

Open Network Linux



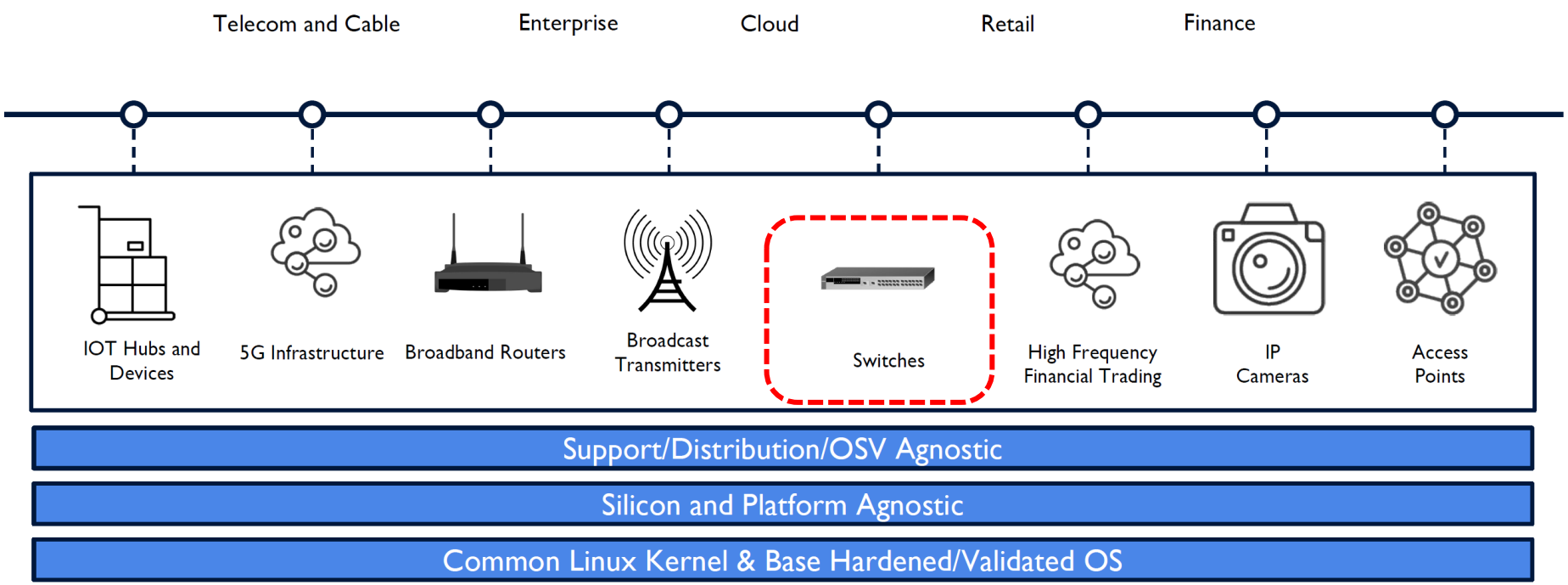
http://www.opennetlinux.org/

https://github.com/opencomputeproject/OpenNetworkLinux



Introducing Network Grade Linux (NGL)

NGL: A COMMON BASE OS SERVING AS AN UPSTREAM FOR NOSes



THE LINUX FOUNDATION

Network Grade Linux



Mion, the initial project of Network Grade Linux, is the Irish word for 'mini'. As an adjective it is also used for small, fine and detailed.

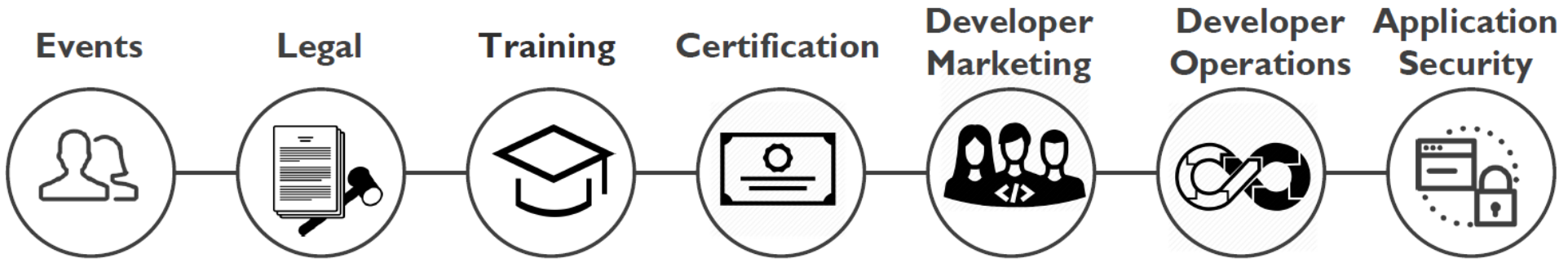
The logo design is based on the atomic configuration of the Helium element, which has a very stable arrangement. Helium is the only element that does not solidify under ordinary pressures and remains a liquid even at absolute zero.

Mion is a backronym for Mini Infrastructure Operating System for Networks

THE LINUX FOUNDATION

- Network Industry 共通の base OS
  - DENT, SONiC, Stratum, 商用NOS等のUpstream
- Yocto Project を利用したセキュアで堅牢なOS
  - AGL(Automotive Grade Linux) や CGL (Carrier Grade Linux) にも利用されている
- CI/CDを考慮した設計や、開発者向けのドキュメントを豊富に用意

Building large, sustainable ecosystems requires collective resources



Events

We gathered over **60,000 attendees** from over **11,000 organizations** across **113 countries** in 2019

Legal

We **manage IP for the worlds most important tech** and have some of the worlds top source legal team in house

Training

We have **trained millions of students** through free and paid online training, online skills certification, and on site e-learning

Certification

We have designed and implemented both software and hardware **testing and certification programs**

Developer Marketing

We have the **largest share of voice of any open source foundation** and a proven method to build large scale developer programs

Developer Operations

We **host the infrastructure that develops the worlds largest software communities** and provide release mgmts., IT ops and support

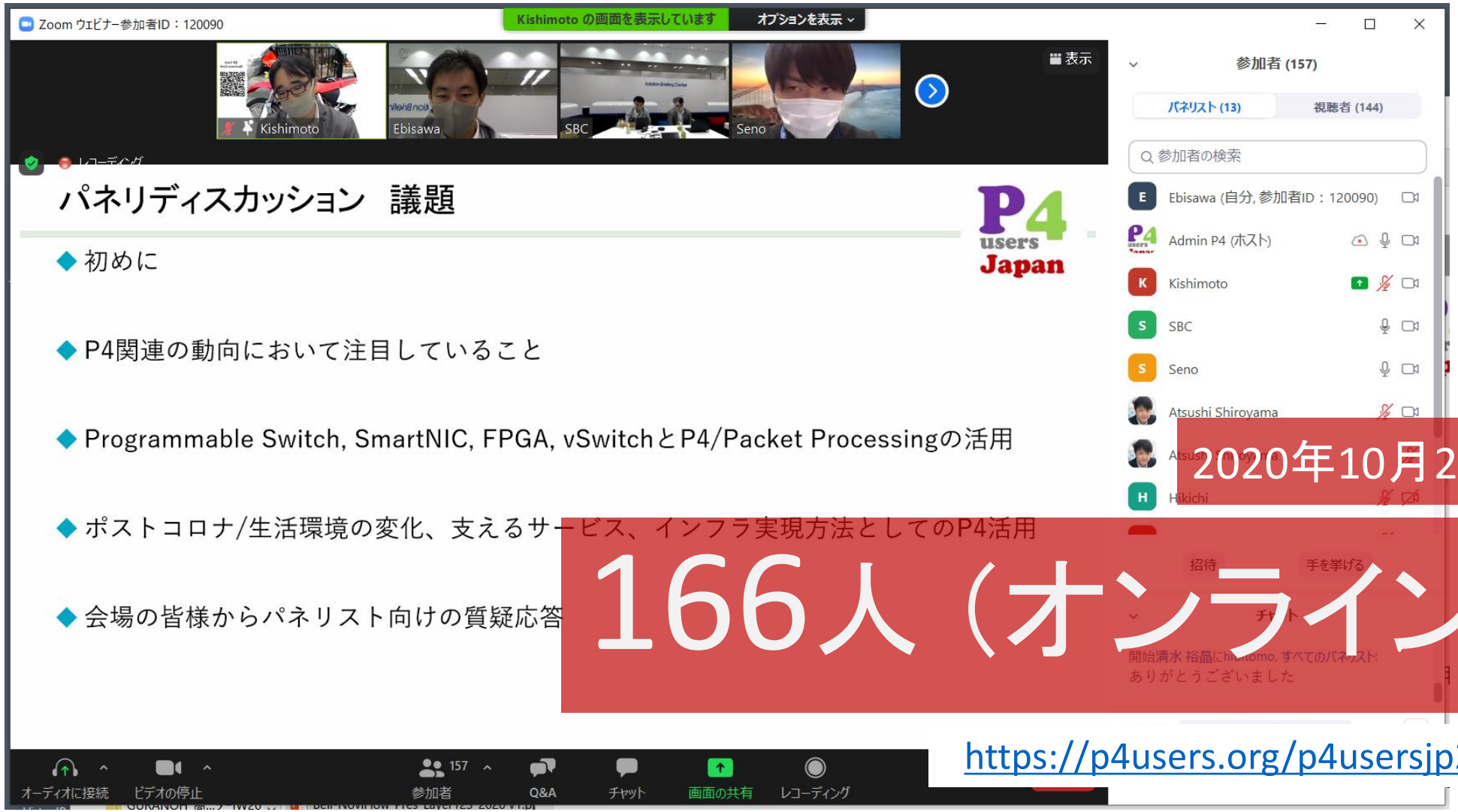
Application Security

In addition to massive peer review, **our projects our regularly audited and pen tested.** We offer bug bounties, dependency analysis, and code scanning.

THE LINUX FOUNDATION

Network Grade Linux をサポートするメンバー企業を募集しています！

日本 P4 ユーザ会（2020）



2020年10月22日

166人（オンライン）

https://p4users.org/p4usersjp2020/

LIVEセッションタイムテーブル

2020年10月22日(木曜日)

No.	時間	タイトル＆発表者
0	09:30-09:35	**イベント参加についてご案内**
1	09:35-09:40	**開催あいさつ** ザケル　ハディ(Zaker Hadi) 日本P4ユーザ会 会長(P4 Users Japan Chairman)
2	09:40-10:10	**P4アップデート** 海老澤 健太郎(Kentaro Ebisawa) トヨタ自動車株式会社(Toyota Motor Corporation)
3	10:10-10:40	**In-band Network Telemetry を活用した 高精度ネットワークモニタリングシステム HANMOC の開発** 妹尾 龍馬(Ryuma Seno) NTTネットワークサービス研究所(NTT Network Service Systems Laboratories)
4	10:40-11:10	**5G利用企業の多様なニーズを「プログラマブル」で叶える次世代コアネットワーク ～マルチHWで動作可能な5GC-UPFの実現～** 白山 篤志(Atsushi Shiroyama) 日本電気通信システム株式会社(NEC Communication System,Ltd.) 西堀 一幹(Chikara Nishibori) 株式会社iD(iD Corporation)
5	11:10-11:40	**パネルディスカッション** テーマ： ・**P4/データプレーンプログラマビリティ、第二章。 The Latest動向と今後の可能性** モデレータ： ・岸本 貴之(APRESIA Systems株式会社) パネリスト： ・海老澤 健太郎(トヨタ自動車株式会社) ・妹尾 龍馬(NTTネットワークサービス研究所) ・白山 篤志(日本電気通信システム株式会社) ・西堀 一幹(株式会社iD) ・あなた(オンライン参加者からの質問を受け付けます)
7	11:50-12:25	**P4ユースケース発表** **1.mobile SRv6(Wedge)(8min)** トヨタ自動車株式会社 海老澤 健太郎 Toyota Motor Corporation, Kentaro Ebisawa **2.SR + INT (FPGA NIC + Wedge)(8min)** 株式会社マクニカ アルティマ カンパニー 引地 祐介 MACNICA, Inc. ALTIMA Company Yusuke Hikichi **3.P4 コンパイラを使って、FPGA に外部の検索エンジンIPを取り込んでみた(10min)** 長瀬産業株式会社 西沢 正登 NAGASE & CO., LTD. Masato Nishizawa 株式会社iD 寺尾 貴明 iD Corporation Takaaki Terao **4.Registerを使ったトラフィック解析の例：Netflow (Wedge)(8min)** APRESIA Systems株式会社 桑田 斉 APRESIA Systems, Ltd. Hitoshi Kuwata
8	12:25-12:50	**ユースケースディスカッション** テーマ： ・**2020年の新しいP4のユースケースとこんなことできたらいいな** モデレータ： ・清水 裕晶(株式会社マクニカ アルティマ カンパニー) パネリスト： ・西沢 正登(長瀬産業株式会社) ・寺尾 貴明(株式会社iD) ・海老澤 健太郎(トヨタ自動車株式会社) ・引地 祐介(株式会社マクニカ アルティマ カンパニー) ・桑田 斉(APRESIA Systems株式会社) ・あなた(オンライン参加者からの質問を受け付けます)

https://p4users.org/p4usersjp2020/

録画セッション

No.	カテゴリ	発表者
1	事例	タイトル： **・IP輻輳の発側制御コンセプト** 概要： 「無駄なトラヒックをネットワークの入り口で破棄する」このコンセプトと実現方法について説明します。 こちらは自律分散型でありながら、全体最適化ができる手法であり、ぜひご覧ください！ プレゼンタ： エヌ・ティ・ティ・コムウェア株式会社 渡邉 みのり NTT COMWARE Corporation Minori Watanabe 動画・資料リンク： PDF：IP輻輳の発側制御コンセプト.pdf 録画デモ：https://www.youtube.com/watch?v=JMaKxr8TZb4 ※PDFと録画デモを合わせてご覧ください
2	事例	タイトル： **NOSもNOSに挑戦（Build the NOS by NOS）** 概要： 本セッションはNOS（Net One Systems）が取り組んでいるユースケースを2点紹介します。 1つめはP4の柔軟性を活用したトラフィックデータのテレメトリを実装した独自INTのユースケースです。 2つめはP4のデータプレーンに独自で実装したマネージメントプレーン（Network Operating Systems）を組み合わせたL1スイッチを作成したユースケースになります。 プレゼンタ： ネットワンシステムズ株式会社 スハルトノ リオスナタ Net One Systems Co., Ltd. Ryosunata Suhartono 動画・資料リンク： 動画：https://www.youtube.com/watch?v=ARqfOlvXBkE
3	製品・スイッチ	タイトル： **P4のP4P** 概要： P4のP4P(Product, Price, Place, Promotion)を当社の切り口で考察してみました。 特にProduct面（ユースケース）とPlace面（サポート）について重点的にお話いたします。 プレゼンタ： APRESIA Systems株式会社 水谷 渉 APRESIA Systems, Ltd. Wataru Mizutani 動画・資料リンク： 動画：https://www.youtube.com/watch?v=gmP0BgzeGT8
4	製品・スイッチ	タイトル： **Arista7170ユースケースの紹介（Introduction Arista7170 Usecase）** 概要： Arista7170シリーズはBarefoot Tofinoを使ったプロダクトです。 本セッションではAristaにおけるBarefootTofinoの実装例/アーキテクチャーを紹介し、 また様々なユースケースに対応できるoff-the-shelfプロファイルの概要に関して説明したいと思います。 プレゼンタ： アリスタネットワークスジャパン合同会社 土屋 師子生 Arista Networks Japan Limited. Shishio Tsuchiya 動画・資料リンク： 動画：https://www.youtube.com/watch?v=p4Xa19jqs_4
5	製品・NIC	タイトル： **P4 FPGA/SW を活用した 新しいアクセラレータ・インターコネクトと INT** 概要： サーバー（FPGA搭載）とスイッチ（P4 Tofino搭載）の組み合わせで、SRv6 動かしてみました！ 更に、この組み合わせ全体で、INT 情報からの遅延計算も行っております！ぜひ、ご覧ください。 プレゼンタ： 株式会社マクニカ　アルティマ カンパニー 引地 祐介 MACNICA, Inc. ALTIMA Company Yusuke Hikichi 動画・資料リンク： 動画：https://www.youtube.com/watch?v=QKXyXmbBlrE
6	製品・チップ	タイトル： **プログラマブル・パケット処理の未来** 概要： Barefootがインテルに買収されて一年。 インテルBarefoot CTOがP4言語を用いたプログラマブル・パケット処理の10年後の未来について、 Tofino、Smart NIC、FPGAに触れながら、いくつかの事例を交えてご紹介します。 プレゼンタ： インテル株式会社 Changhoon Kim 動画・資料リンク： 動画：https://www.youtube.com/watch?v=-A44d_aH27A
7	製品・チップ	タイトル： 次世代SDNにおけるP4の役割 概要： ONF の Timone Sloane からの動画です！ONF の各種プロジェクト、 例えば SEBA、VOLTHA、AETHER、合わせて ONOS、Stratum の現状と P4 の関わりがわかります。 プレゼンタ： ONF Timon Sloane 動画・資料リンク： 動画：https://www

https://p4users.org/p4usersjp2020/

ホワイトボックス・スイッチ「今後への期待」（コミュニティ状況）

https://p4users.org/





https://p4users-jp.slack.com/

!! ONLINE !!

BAREFOOT ACADEMY™

9月28日～10月30日 | 全10回 | P4の基本 ～ Tofino Native Architecture 詳細

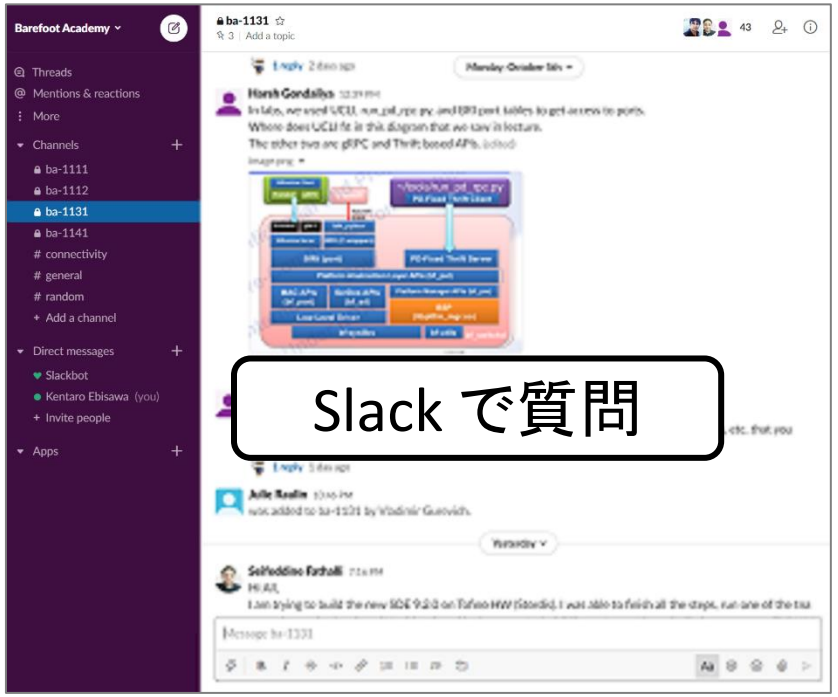
https://www.barefootnetworks.com/barefoot-academy/



講義＆質疑応答
（約40名参加・日本から3名）

Online Hands On

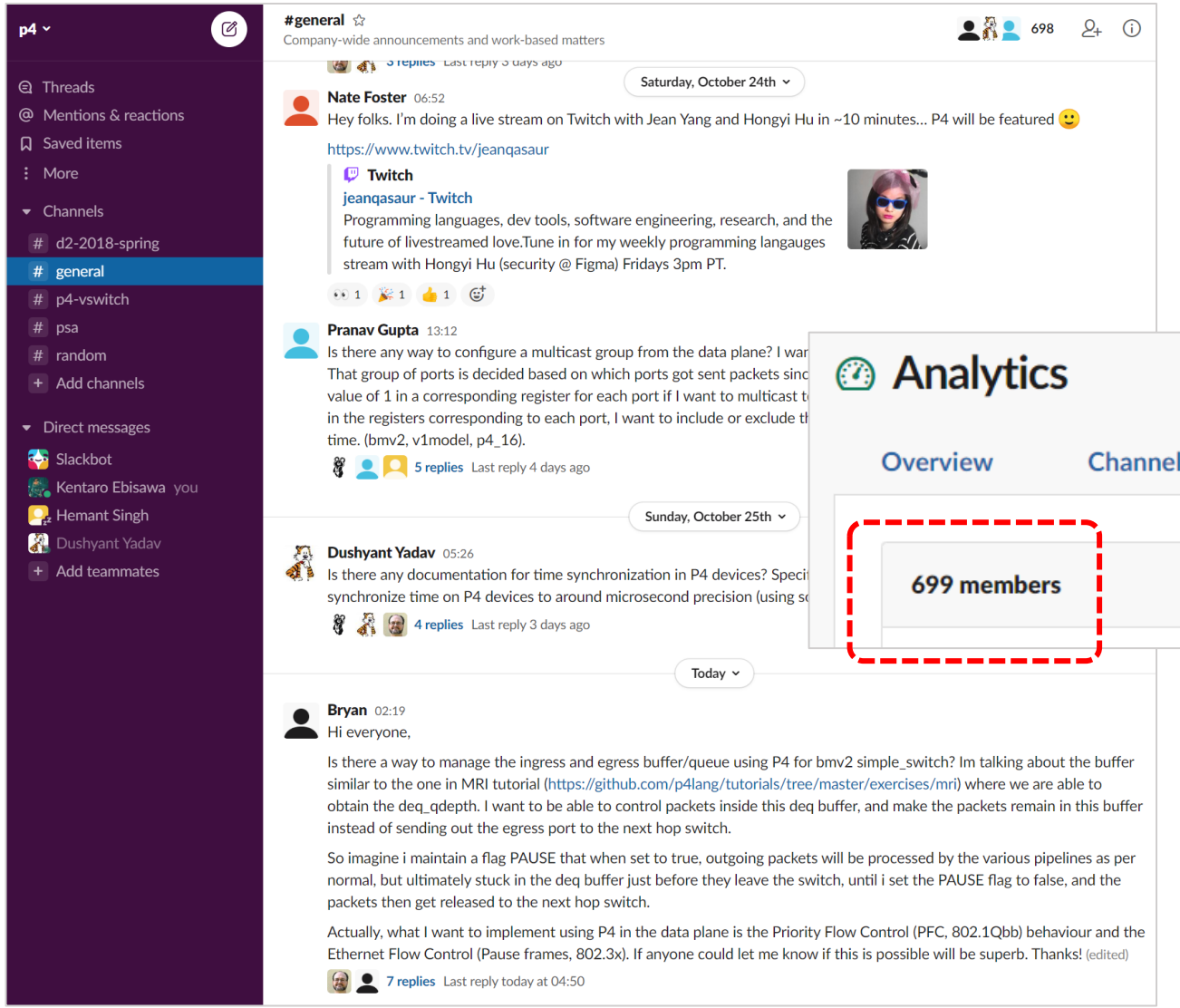




Slack で質問

データプレーンプログラミングの裾野拡大中

p4-lang.slack.com



最近の話題

P4Runtime Client (Go!)
Digest, Register Read
Const table entry
Clone Packet
ARP packet handling
Timer and Aging
ONOS
Packet Test Framework
P4_16 / P4_14 conversion
Workshop and Events

ホワイトボックス・スイッチが広げる「ネットワークの可能性」

データプレーン プログラミング「P4の動向」

「In-Network Computing の研究動向」

ホワイトボックス・スイッチ「今後への期待」

実現できる「アーキテクチャ」や「ユースケース」は無限大！
ホワイトボックス・スイッチの新しい可能性を探り実装していきましょう